Sharp Scalar Elliptic Regularity and Certified Degenerate Response

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Abstract

We give a scalar-potential extension of the constrained-gradient calibration of Agazzi et al. to real div–curl pairs in every dimension. It yields the exact L^p norm $\max\{p-1,(p-1)^{-1}\}$ of the real-vector Hodge reflection and the sharp open gradient-estimate interval for measurable symmetric scalar elliptic equations, together with the attained sharp local weak endpoint for homogeneous energy solutions and their normalized fluxes. The same calibration, integrated against damped logarithmic weights, proves sharp local regularity under exponentially integrable distortion: if $R^2 \le KE\cdot B$ and $\exp(\tau K)$ is locally integrable, then $(E\cdot B)\log^{\beta}(\mathrm{e}+E\cdot B)$ and $(|E|^2+|B|^2)\log^{\beta-1}(\mathrm{e}+|E|+|B|)$ are locally integrable for every $0<\beta<\tau$. Explicit examples exclude the logarithmic strong endpoint and every general power gain in the degenerate class. We then establish periodic finite-energy solvability, density in the actual anisotropic primal and dual energy spaces, certified rational approximation, and variational homogenization when the coefficient and its inverse have a supplied exponential moment. On a bounded C^2 domain, smooth macroscopic solutions admit an $O(\sqrt{\delta})$ L^2 and weighted corrector estimate. A logarithmic modulus also controls effective-response changes under small-region, relatively comparable coefficient edits. The computational theorem uses expanded coefficient data with certified integral error and pays for their acquisition; it is a termination theorem, not a uniform polynomial-time bound. The critical positive div–curl product and the hypercircle principle are identified as prior machinery.

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1 Introduction and main results

An elliptic equation couples two different constraints. A field is a gradient, and its constitutive flux is divergence free. Neither constraint controls large values by itself. The useful cancellation occurs when they are paired. This paper develops one calibration of that pairing in two directions: sharp power estimates for uniformly elliptic equations, and sharp logarithmic estimates when the ellipticity degenerates.

The geometric input is unusually elementary. Project a scalar potential onto the convex set of potentials whose gradients remain in a prescribed ball. Where the projected gradient differs from the original one, it reaches the boundary of that ball. Translation of the admissible set gives a second, integrated positivity property. These two facts produce a calibration for a gradient and an arbitrary divergence-free vector field. They require neither a planar stream function nor a vector-valued gradient constraint.

1.1 Scope, notation and the principal statements

Throughout $d \geq 2$, $Q = \mathbb{R}^d/\mathbb{Z}^d$ has volume one, and vector norms are Euclidean. The equations have a scalar unknown. A coefficient matrix acts on its gradient; it does not make the equation an elliptic system. For a measurable set D of finite positive volume, $\langle f \rangle_D = |D|^{-1} \int_D f$. All differential constraints are distributional unless more regularity is specified. On Q, define

$$\widehat{Pf}(m) = \begin{cases} mm^T \widehat{f}(m)/|m|^2, & m \neq 0, \\ 0, & m = 0, \end{cases} \qquad H = 2P - I, \qquad C_p = \max\{p-1, (p-1)^{-1}\}. \tag{1.1}$$

On $\mathbb{R}^d$ the same multiplier is used away from the origin. Thus P projects onto gradients with zero mean on the torus, and $H^2 = I$. In particular, H acts as $-I$ on constant fields.

Theorem 1.1 (Sharp reflection and scalar interval)**.** *For $1 < p < \infty$, on both Q and $\mathbb{R}^d$,*

$$\|H\|_{L^p(\mathbb{R}^d;\mathbb{R}^d) \to L^p(\mathbb{R}^d;\mathbb{R}^d)} = C_p,$$

with Q substituted for $\mathbb{R}^d$ in the periodic assertion. Let A be measurable, real symmetric, and $\lambda I \leq A \leq \Lambda I$, where $0 < \lambda \leq \Lambda < \infty$. Put $s = \sqrt{\Lambda/\lambda}$. For

$$\frac{2s}{s+1} < p < \frac{2s}{s-1}, \tag{1.2}$$

the right endpoint being infinity if $s = 1$, every $F \in L^p(Q;\mathbb{R}^d)$ has a unique mean-zero $u \in W^{1,p}(Q)$ satisfying $\operatorname{div}(A\nabla u) = \operatorname{div} F$. Moreover,

$$\|\nabla u\|_p \leq \frac{C_p(1+C_p)}{\lambda[(s+1) - (s-1)C_p]} \|F\|_p. \tag{1.3}$$

The open interval is maximal uniformly over this coefficient class. The displayed constant is sufficient, not asserted optimal.

The Euclidean elliptic assertion concerns the unique L^p gradient and a homogeneous Sobolev potential modulo constants. There is no general-domain boundary assertion. For $p < 2$, the periodic solution is a $W^{1,p}$ solution; it need not have finite quadratic energy for arbitrary L^p data. Agreement with the energy solution on the overlap of the data spaces is proved in Section 3.

Theorem 1.2 (Sharp local weak endpoint)**.** *Let $0 < \lambda < \Lambda$, $s = \sqrt{\Lambda/\lambda} > 1$ and $p_+ = 2s/(s-1)$. Let A be measurable and real symmetric on $B_{4r}(x_0)$, with $\lambda I \le A \le \Lambda I$. If $u \in W^{1,2}(B_{4r}(x_0))$ satisfies $\operatorname{div}(A\nabla u) = 0$, then, for each of the fields*

$$g = \nabla u \quad \text{and} \quad g = \frac{A\nabla u}{\sqrt{\lambda\Lambda}},$$

one has

$$\sup_{t>0} t \left(\frac{|\{x \in B_r(x_0) : |g(x)| > t\}|}{|B_r|} \right)^{1/p_+} \le C(d,s) \left(\fint_{B_{4r}(x_0)} |\nabla u|^2 \, \mathrm{d}x \right)^{1/2}. \tag{1.4}$$

No larger weak exponent, or finite Lorentz index at p_+, holds uniformly over this class.

This local homogeneous assertion is distinct from the global forced strong estimates in (1.2). It makes no claim about the global forced endpoint operator or arbitrary boundary data. The flux normalization is essential for the stated constant. Section 3.5 gives the complete localization and tail proof, without a critical-product import, and the homogeneous uniformly convex nonlinear corollary. If $s = 1$, the coefficient is λI and the corresponding local L^∞ bound is the usual harmonic estimate.

For the degenerate statement define, for a pair E, B,

$$J = E \cdot B, \qquad R = \frac{|E+B| + |E-B|}{2}, \qquad V = (|E|^2 + |B|^2)^{1/2}. \tag{1.5}$$

Theorem 1.3 (Sharp logarithmic regularity)**.** *Let $E, B \in L^1_{\mathrm{loc}}(\Omega; \mathbb{R}^d)$, where $\Omega \subset \mathbb{R}^d$ is open. Suppose*

$$\operatorname{curl} E = 0, \quad \operatorname{div} B = 0, \quad 0 \le J \in L^1_{\mathrm{loc}}, \quad R^2 \le KJ, \quad K \ge 1, \quad \mathrm{e}^{\tau K} \in L^1_{\mathrm{loc}}$$

for a finite almost-everywhere K and $\tau > 0$. For every $0 < \beta < \tau$,

$$J \log^\beta(\mathrm{e} + J), \qquad V^2 \log^{\beta-1}(\mathrm{e} + V) \quad \textit{belong to } L^1_{\mathrm{loc}}(\Omega). \tag{1.6}$$

More precisely, for concentric $B_r \subset B_{2r} \Subset \Omega$, set

$$q = 2 - \frac{1}{d+1}, \quad m = 2d+1, \quad E_0 = \langle J \rangle_{B_{2r}}, \quad M_0 = \langle \mathrm{e}^{\tau K} \rangle_{B_{2r}}, \quad A_0^2 = \frac{2m}{\mathrm{e}\tau} E_0 M_0^{1/m}.$$

If $E_0 > 0$, then

$$\fint_{B_r} \left[\frac{J}{A_0^2} \log^\beta \left(\mathrm{e} + \frac{J}{A_0^2} \right) + \frac{V^2}{A_0^2} \log^{\beta-1} \left(\mathrm{e} + \frac{V}{A_0} \right) \right] \mathrm{d}x \le C(d, \beta, \tau)(1 + M_0). \tag{1.7}$$

If $E_0 = 0$, both fields vanish almost everywhere on B_{2r}. The endpoint $\beta = \tau$ fails in general, and even finiteness of every exponential moment of K does not imply any $L^{2+\delta}$ gain.

The threshold is independent of dimension. The localization constants in (1.7) are not asserted to be dimension independent or numerically evaluated. The inequality $R^2 \le KJ$ is equivalent to $V^2 \le (K + K^{-1})J$. Section 4 proves this equivalence and the theorem without assuming separate L^2 integrability of the fields.

The final application concerns a periodic matrix A which is symmetric positive definite almost everywhere, with

$$K = \max\{1, \|A\|_{\mathrm{op}}, \|A^{-1}\|_{\mathrm{op}}\}, \qquad \int_Q \mathrm{e}^{\tau K} \, \mathrm{d}x \le M < \infty. \tag{1.8}$$

The units are fixed: replacing $A(x)$ by an x-dependent scalar normalization changes the equation.

Theorem 1.4 (Degenerate periodic response)**.** *Under* (1.8)*, for every* $\boldsymbol{e} \in \mathbb{R}^d$ *and bounded* $F : Q \to \mathbb{R}^d$*, there is a unique finite-energy periodic response*

$$E = \boldsymbol{e} + \nabla u, \quad \langle u \rangle_Q = 0, \quad B = AE - F, \quad \operatorname{div} B = 0.$$

Here $u \in W^{1,q}(Q)$ *for every* $1 < q < 2$*. Smooth admissible gradients and smooth divergence-free fields are dense in their respective* A *and* A^{-1} *energy spaces. Consequently the primal and dual energies give exact squared-error certificates.*

If A *is supplied by a finite program producing expanded rational step matrices with certified* L^1 *operator-norm errors, and rational* $\tau, M, \boldsymbol{e}, F$ *are supplied as specified in Section 7, a deterministic rational algorithm terminates at every prescribed positive tolerance. It constructs inverse-coefficient data and compatible fields rather than querying an inverse or projection oracle. Its costs include the coefficient program, expanded grids and integer arithmetic; no uniform polynomial bound in succinct input length or precision bits is claimed.*

For $F = 0$*, the resulting tensor* $A_*\boldsymbol{e} = \langle B \rangle_Q$ *is the variational periodic homogenized tensor, with strong-*L^2 *Gamma convergence on bounded Lipschitz domains with zero trace. For a bounded* C^2 *domain and* $u_0 \in C^2(\overline{\Omega}) \cap H^1_0(\Omega)$*, the loading* $f = -\operatorname{div}(A_* \nabla u_0)$ *gives the quantitative estimate*

$$\|u_\delta - u_0\|_{L^2(\Omega)} \le C(d, \tau, M, \Omega)\sqrt{\delta}\|u_0\|_{W^{2,\infty}(\Omega)}.$$

The precise weighted corrected-gradient statement and both domain conventions are in Theorem 8.3. Corollary 8.1 gives a logarithmic effective-tensor modulus for edits on a small set, assuming bounded relative comparability and the stated moments for both media.

1.2 Relation to previous work

Meyers's classical higher-integrability theorem [14, Theorem 2] gives a coefficient-class neighbourhood of the energy exponent. The sharp planar interval is already part of Beltrami-operator theory [5]; see also the remark following [12, equation (11.10)]. The contribution developed here is the explicit all-dimensional real-vector calibration and its scalar, logarithmic and certified periodic consequences. The radial obstruction to a larger uniform interval and its dimensional descent are also classical [14, Section 5].

The planar weak endpoint is also established. In the symmetric normalization $s^{-1}I \le A \le sI$, its exponent is $2s/(s-1)$. Nesi–Palombaro–Ponsiglione [15, p. 617] record this endpoint and the precedent of Leonetti–Nesi [13]. The all-dimensional local proof here develops the calibration's tail inequality; it does not claim the planar conclusion as new or assert historical firstness in higher dimension.

Agazzi et al. [1], Theorem 1 and Corollary 2, prove the planar calibration and the sharp Beurling–Ahlfors norm. Their Lemma 4 supplies the scalar constrained-gradient projection; their introduction explicitly announces higher-dimensional and elliptic developments. Here the scalar ball constraint is written out in arbitrary dimension, with divergence replacing the planar null-Lagrangian pairing. The subsequent power and logarithmic arguments are developed for these real-vector pairs. We make no claim to priority over announced or subsequent developments. The planar integrand predates that proof: its origins include Burkholder's differential-subordination construction and Šverák's rank-one-convex example [9, 18]. The power majorants and the calibration-to-integral framework belong to this earlier tradition; see also Baernstein–Montgomery-Smith [6].

Iwaniec and Sbordone [12] formulate the real-vector reflection question, up to its harmless overall sign convention, in Conjecture 9.1 and connect it to the interval in their equation (11.10). Their Conjecture 1.1 concerns a broader distortion programme. The results here cover the scalar real-vector setting stated above. They do not assert a norm for every degree of differential form or resolve a general elliptic-system conjecture.

Prior machinery	Use and development here	Scope retained
Scalar projection and planar calibration [1]	All-dimensional scalar-potential calibration; full mean terms	Real vectors, Euclidean norm
Planar finite-distortion regularity [4]	Damped div–curl proof and cylindrical sharpness examples	Scalar gradient and divergence constraints
Planar weak endpoint [15]	Localized calibration tail argument in every dimension	Homogeneous energy solutions; normalized flux
Critical positive product [17]	Justifies localization and eliminates the energy-density residual	Critical logarithmic classes; positivity is essential
Weighted-density theory [3]	Separate primal and dual proofs in the matrix energies	No substitution by a stronger scalar-weight norm
Hypercircle and certified homogenization [16, 19]	Certified coefficient and inverse uncertainty without an ellipticity gap	Represented input; paid finite computation
Quantitative corrector methodology [11, 7]	Weighted potentials, boundary strip and variational residual	Periodic coefficients; smooth macroscopic solution

The sharp planar logarithmic threshold and radial endpoint method are established in Astala–Gill–Rohde–Saksman [4]. The local argument below uses scalar and antisymmetric regularized potentials [10], and an explicitly stated critical-product theorem [17]. These imports are kept separate from the calibration. In the application, primal–dual certification is classical, and degenerate numerical analysis already includes Muckenhoupt-weight settings [2]. Our hypothesis does not impose that condition. The homogenization proof uses classical variational methodology [8]; it supplies the particular degenerate admissibility and approximation steps needed here. The square-root rate uses established corrector methodology [11, 7], with a separate proof under the periodic exponential-moment contract and explicit macroscopic smoothness.

1.3 How the proofs fit together

For power estimates, integrating the calibration in its threshold produces a Burkholder-type expression. A scalar majorization then bounds the reflection. For logarithmic estimates, the threshold weight is damped before integration. This prevents assuming the very integrability one is trying to prove. The leading coefficients of the two threshold integrals are $1/\beta$ and $1/2$; they permit absorption precisely below $\beta = \tau$.

At the power endpoint, keep the calibration threshold instead of integrating it out. The exact coefficient in the localized distortion inequality gives a differential inequality for $\int (r-t)_+^2$. Subcritical bootstrap makes the cutoff errors decay faster than its critical power. Integration yields the weak distribution bound, although the strong endpoint still fails.

For periodic response, positivity provides a different benefit. A residual orthogonal to all smooth admissible fields would create a positive mean-zero div–curl pair. The critical-product theorem makes its integral zero, so the residual vanishes. This proves density in both actual matrix energy spaces. Compatible primal and dual approximations can then certify the continuum response even when both A and A^{-1} are unbounded. For the finite-scale rate, a boundary cutoff makes the first-order corrector admissible. Weighted potential estimates control its boundary cost, and an antisymmetric flux potential controls its residual. Periodic density on a containing cube extends this argument to the maximal variational domain.

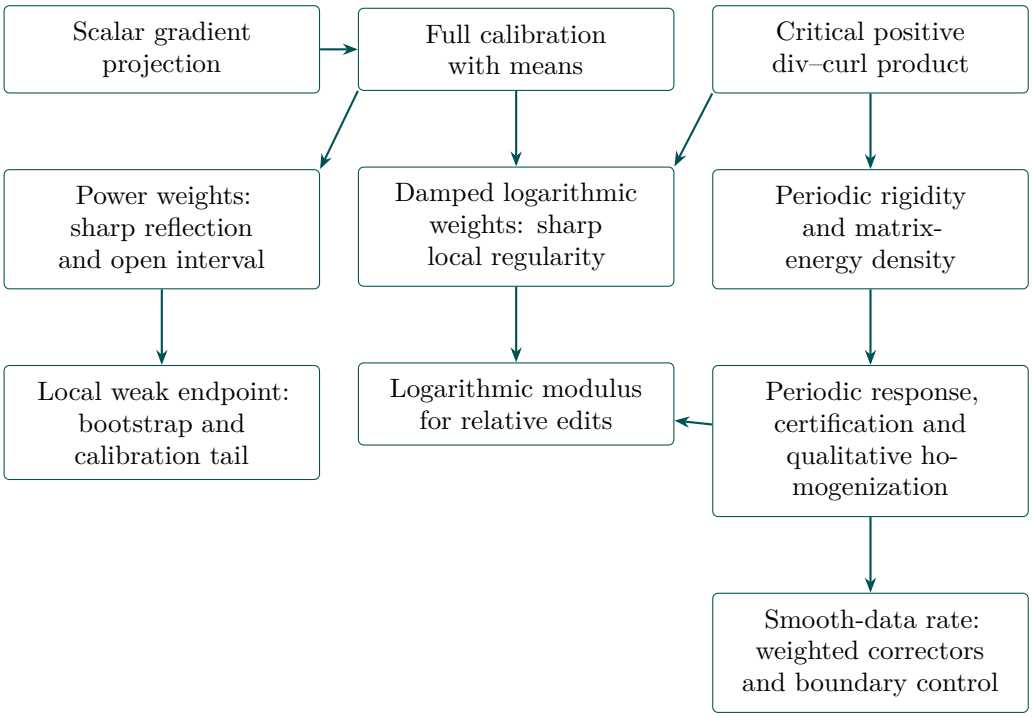


Figure 1: Three uses of the calibration. The local weak endpoint uses ordinary L^2 products; the critical positive-product theorem enters the logarithmic and density branches. The smooth-data rate uses weighted corrector potentials and density, independently of the finite algorithm.

2 The calibration and the exact reflection norm

2.1 Projection onto a scalar gradient constraint

The next argument is the ball-constrained version of [1, Lemma 4]. We include the dimension-dependent details: the admissible object is a scalar potential on Q, and its gradient lies in a Euclidean ball in $\mathbb{R}^d$.

Lemma 2.1 (Constrained scalar projection)**.** *Let $t > 0$, $|a_0| \le t$, and $\eta \in W^{1,\infty}_{\mathrm{per}}(Q)$. There is $\zeta \in W^{1,\infty}_{\mathrm{per}}(Q)$ such that, with $a = a_0 + \nabla\eta$ and $z = a_0 + \nabla\zeta$,*

$$|z| \le t, \qquad z = a \text{ or } |z| = t \quad \text{a.e.}, \qquad \int_Q z \cdot (a - z)\,\mathrm{d}x \ge 0. \tag{2.1}$$

Proof. Consider the convex subset of $L^2(Q)$

$$\mathcal{C} = \{\xi \in W^{1,\infty}_{\mathrm{per}}(Q) : |a_0 + \nabla\xi| \le t \text{ a.e.}\}.$$

It contains constants. It is closed in L^2: an L^2-convergent sequence has uniformly bounded gradients, bounded means, and hence uniformly bounded Lipschitz representatives. A subsequence converges uniformly, and weak-* convergence of its gradients preserves the convex ball constraint. The Hilbert projection ζ of η therefore exists uniquely. Put $h = \eta - \zeta$. Its characterization is

$$\int_Q h(\xi - \zeta)\,\mathrm{d}x \le 0 \qquad (\xi \in \mathcal{C}). \tag{2.2}$$

Write $w = a_0 \cdot x + \zeta$. On $\{h = 0\}$, the scalar Sobolev level-set property gives $\nabla h = 0$ almost everywhere, so $z = a$. At a differentiability point x_0 of w where $h(x_0) > 0$, suppose

$|\nabla w(x_0)| < t$. Choose $\delta > 0$ with $|\nabla w(x_0)| + 2\delta < t$. For a sufficiently small ball $B_r(x_0)$, contained in a coordinate chart of Q, $h > 0$ and

$$|w(x_0 + y) - w(x_0) - \nabla w(x_0) \cdot y| \le \delta |y|.$$

The cone $c(x) = w(x_0) + \epsilon - t|x - x_0|$, with $0 < \epsilon < \delta r$, lies below w on the boundary and above it at x_0. Replacing w by $\max(w, c)$ in that ball changes ζ by the nonzero compactly supported function $(c - w)_+$. Both gradients satisfy the ball constraint. This contradicts (2.2). For $h(x_0) < 0$, use the minimum with the upward cone $w(x_0) - \epsilon + t|x - x_0|$. Rademacher's theorem proves the active-set assertion almost everywhere. This cone construction works in every dimension.

Finally $\zeta(\cdot \pm s e_i) \in \mathcal{C}$. Add their two variational inequalities and change variables to obtain

$$\int_Q [h(x + s e_i) - h(x)][\zeta(x + s e_i) - \zeta(x)] \, \mathrm{d}x \ge 0.$$

Divide by s^2 and let $s \to 0$, using $W^{1,2}$ difference quotients. Summing over $i = 1, \dots, d$ gives $\int_Q \nabla\zeta \cdot \nabla h \ge 0$. The constant term contributes $a_0 \cdot \int_Q \nabla h = 0$, which proves the last assertion of (2.1). □

2.2 The full mean-sensitive inequality

For $a, b \in \mathbb{R}^d$ set

$$\alpha = \tfrac{1}{2}|a + b|, \quad \gamma = \tfrac{1}{2}|a - b|, r = \alpha + \gamma, \quad D = a \cdot b = \alpha^2 - \gamma^2.$$

Use r for a generic pair and R for the physical pair in (1.5). Define

$$L_t(a, b) = \begin{cases} D, & r \le t, \\ 2t\alpha - t^2, & r > t. \end{cases} \tag{2.3}$$

The definitions agree at $r = t$.

Lemma 2.2 (Algebraic support formula). *Let $Z_t(a) = \{z : |z| = t\}$, augmented by $\{a\}$ when $|a| \le t$. Then*

$$L_t(a, b) = \sup_{z \in Z_t(a)} \{z \cdot b + z \cdot (a - z)\}. \tag{2.4}$$

Moreover $L_t \ge t|a + b| - t^2$, and L_t is globally Lipschitz in (a, b), with constant at most $4t$.

Proof. On the sphere the supremum is $t|a + b| - t^2$; at $z = a$ the value is D. The comparison follows from

$$D - (2t\alpha - t^2) = (t - \alpha - \gamma)(t - \alpha + \gamma), \qquad |\alpha - \gamma| \le |a| \le \alpha + \gamma.$$

These inequalities decide the maximum whenever the extra point is admissible, and give (2.4) in the other cases as well. On $r < t$, $|a|, |b| \le t$, so the derivatives of D are bounded by t in each variable. On $r > t$ the assertion follows directly from $t|a + b| - t^2$. Continuity across the interface gives the stated global Lipschitz bound, by restriction to line segments. □

Proposition 2.3 (Calibration with means). *If $a = a_0 + \nabla\eta$, $\eta \in W^{1,\infty}_{\mathrm{per}}(Q)$, and $b \in L^1(Q; \mathbb{R}^d)$ is divergence free with mean b_0, then*

$$\int_Q L_t(a, b) \, \mathrm{d}x \ge L_t(a_0, b_0). \tag{2.5}$$

Proof. If $r(a_0, b_0) > t$, Lemma 2.2 and convexity of the norm give the result immediately. Otherwise $|a_0| \le t$. Use Lemma 2.1 and (2.4). Since $z \in Z_t(a)$ almost everywhere,

$$\int_Q L_t(a,b)\,\mathrm{d}x \ge \int_Q z \cdot b\,\mathrm{d}x + \int_Q z \cdot (a-z)\,\mathrm{d}x \ge a_0 \cdot b_0 = L_t(a_0, b_0).$$

For the first equality, a periodic Lipschitz potential may be tested against an L^1 divergence-free field: smooth its potential, use uniformly bounded gradients and almost-everywhere convergence, and apply dominated convergence. This accounts for both means. □

Corollary 2.4 (Compact L^1 version)**.** *If $a, b \in L^1(\mathbb{R}^d; \mathbb{R}^d)$ have compact support,* $\operatorname{curl} a = 0$*, and* $\operatorname{div} b = 0$*, then* $\int_{\mathbb{R}^d} L_t(a,b)\,\mathrm{d}x \ge 0$*.*

Proof. For smooth fields, put their support inside a periodic cube. A compactly supported curl-free field has zero mean: write $a = \nabla v$ and choose the potential constant off a large ball. A compactly supported divergence-free field has zero mean because $\int b_i = \int \operatorname{div}(x_i b) = 0$. Rescale Proposition 2.3 to the cube. For L^1 fields mollify both, preserving the constraints and compact support. The Lipschitz estimate in Lemma 2.2 passes the integrals to the limit. No product integrability is required for this step. □

2.3 Power weights and majorization

Lemma 2.5 (Integral representation)**.** *For $p > 2$ let*

$$\mathcal{B}_p(a,b) = ((p-1)\gamma - \alpha)(\alpha + \gamma)^{p-1}.$$

Then

$$\mathcal{B}_p = \frac{p(p-1)(p-2)}{2} \int_0^\infty t^{p-3}[L_t(a,b) - D]\mathrm{d}t. \tag{2.6}$$

For bounded periodic div–curl pairs, $\int_Q \mathcal{B}_p(a,b) \ge \mathcal{B}_p(a_0, b_0)$. Furthermore, with $c_p = p(1 - 1/p)^{p-1}$,

$$\alpha^p - (p-1)^p \gamma^p \le -c_p \mathcal{B}_p(a,b). \tag{2.7}$$

Proof. The integrand vanishes for $t \ge r$. Direct integration on $(0, r)$ gives

$$-\frac{D r^{p-2}}{p-2} + \frac{2\alpha r^{p-1}}{p-1} - \frac{r^p}{p}.$$

Multiplication by the coefficient in (2.6) yields the stated expression. At $r = 0$ both sides are zero. The bound by a constant times r^p justifies integration in space for bounded fields. Subtract the expression for the means and use $\int_Q D = a_0 \cdot b_0$ and (2.5). Although $L_t - D$ can have either sign, the spatially integrated difference has the required sign at every t; the absolute integrability just established permits Fubini.

For (2.7), divide by r^p if $r > 0$ and set $v = \gamma / r \in [0, 1]$. It is enough to show

$$g(v) = c_p(1 - pv) - (1-v)^p + (p-1)^p v^p \ge 0.$$

Here $g(1/p) = g'(1/p) = 0$, and

$$g''(v) = p(p-1)[(p-1)^p v^{p-2} - (1-v)^{p-2}]$$

changes sign once, at $v_0 = [1 + (p-1)^{p/(p-2)}]^{-1} < 1/p$. On $[v_0, 1]$ the function is convex and its minimum is zero. On $[0, v_0]$ it is concave, so its minimum is at an endpoint. The endpoint v_0 is already nonnegative, while $g(0) = c_p - 1 \ge 0$. For the latter fact, $c_2 = 1$ and

$$\frac{\mathrm{d}}{\mathrm{d}p} \log c_p = \frac{2}{p} + \log(1 - 1/p) \ge \frac{2}{p} - \frac{1}{p-1} \ge 0 \quad (p \ge 2).$$

This proves the scalar majorization. □

For a real trigonometric polynomial f, choose $a = Pf$, $b = -(I-P)f$. Then $a+b = Hf$, $a-b = f$; the means are $a_0 = 0$, $b_0 = -\langle f\rangle_Q$, so $\alpha_0 = \gamma_0$ and $\mathcal{B}_p(a_0, b_0) \geq 0$. Lemma 2.5 therefore proves $\|Hf\|_p \leq (p-1)\|f\|_p$ for $p > 2$. Density extends H to $L^p(Q)$. For $p = 2$, P is orthogonal and H is an isometry. The real L^2 pairing makes H self-adjoint. Duality gives $\|H\|_p = \|H\|_{p'} \leq (p-1)^{-1}$ below two, and also shows that the extensions retain $H^2 = I$.

2.4 Transfer to Euclidean space

First let f be a real Schwartz field with Fourier transform supported in an annulus away from zero. The fields $a = Pf$ and $b = -(I-P)f$ have Schwartz potentials

$$a = \nabla\phi, \qquad b_i = \sum_j \partial_j \Psi_{ij}, \quad \Psi_{ij} = -\Psi_{ji}, \qquad \widehat{\Psi}_{ij}(\xi) = \frac{\xi_j \widehat{b}_i - \xi_i \widehat{b}_j}{2\pi i|\xi|^2}.$$

Let $\chi_R(x) = \chi(x/R)$ with $\chi = 1$ near the origin. The compact fields

$$a_R = \nabla(\chi_R\phi), \qquad (b_R)_i = \sum_j \partial_j(\chi_R \Psi_{ij})$$

are exactly curl free and divergence free. They converge to a, b in every L^p, $1 < p < \infty$. On a periodic cube containing their support, apply the preceding calibration, representation and majorization with zero means. Then let $R \to \infty$. This proves the Euclidean upper bound on this dense class.

For completeness, annular Fourier-supported Schwartz fields are dense in L^p. Approximate first by smooth compactly supported fields; smooth high-frequency truncation converges by the approximate-identity theorem, while a smooth low-frequency cutoff shrunk to zero converges to zero in L^p, $p > 1$, first on such test fields by scaling and then by its uniformly bounded convolution norm. The norm estimate extends H to all of $L^p(\mathbb{R}^d)$. Orthogonality and duality treat the remaining exponents as on Q.

2.5 Sharpness and the order of limits

Fix $p > 2$ in the plane and let $U(z) = z|z|^{-2/p}$ on $\epsilon < |z| < 1$. Choose smooth radial cutoffs, zero for $|z| < \epsilon/2$ and $|z| > 2$, equal to one on the displayed annulus, with the usual scale bounds on their derivatives. Denote the resulting smooth compactly supported map by $U_\epsilon = u_\epsilon + iv_\epsilon$. On the annulus,

$$|\partial_z U| = (1-1/p)|z|^{-2/p}, \qquad |\partial_{\bar{z}} U| = p^{-1}|z|^{-2/p}.$$

The p-th power integrals on both cutoff annuli are bounded independently of ϵ, by rescaling at the inner annulus. The main integrals are proportional to $\log(1/\epsilon)$. Consequently

$$\frac{\|\partial_z U_\epsilon\|_p}{\|\partial_{\bar{z}} U_\epsilon\|_p} \longrightarrow p-1.$$

With $\mathsf{J} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, put $a = \nabla u_\epsilon$, $b = -\mathsf{J}\nabla v_\epsilon$. The elementary Cauchy–Riemann identities give $|a+b| = 2|\partial_z U_\epsilon|$, $|a-b| = 2|\partial_{\bar{z}} U_\epsilon|$. For $f = a-b$, exact differential constraints give $Hf = a+b$. This proves the lower bound in $\mathbb{R}^2$ and, by placing the compact potentials inside a periodic cell and rescaling, on $\mathbb{T}^2$.

On $\mathbb{T}^d$ the same fields may be independent of the last $d-2$ coordinates. In $\mathbb{R}^d$, multiply both planar potentials by a smooth cutoff $\theta(x'/L)$ in the other coordinates. Use the full gradient of the first potential and use $-\mathsf{J}\nabla_{x_1,x_2}$ of the second in the first two components, with the other flux components zero. These fields still satisfy the exact differential constraints. For

fixed ϵ, their extra gradient components have L^p norm $O(L^{-1}L^{(d-2)/p})$, whereas the principal norms scale as $L^{(d-2)/p}$. Let $L \to \infty$ first and then $\epsilon \to 0$. This order avoids a uniform cutoff estimate in a parameter where none is needed. For $p < 2$, self-adjoint duality transfers the lower bound from $p' > 2$. At $p = 2$ it is one. This proves the reflection assertion of Theorem 1.1.

3 Scalar ellipticity and constitutive inversion

3.1 A contraction principle with an error certificate

The reflection has square equal to the identity. Thus a small Lipschitz perturbation of it can be inverted by a fixed-point iteration. The norm proved above determines exactly how small the perturbation must be.

Theorem 3.1 (Lipschitz constitutive inversion)**.** *Let $1 < p < \infty$. Suppose $m(x, z)$ is measurable in x, continuous in z, $m(x, 0) = 0$, and*

$$|m(x, z) - m(x, w)| \le k|z - w|$$

almost everywhere in x, for all $z, w \in \mathbb{R}^d$. If $\vartheta = kC_p < 1$, then, on Q or $\mathbb{R}^d$, every $h \in L^p$ has a unique solution of $Hf - m(x, f) = h$. Starting from $f_0 = 0$,

$$f_{j+1} = H(m(x, f_j) + h)$$

converges in L^p, and

$$\|f\|_p \le \frac{C_p}{1 - \vartheta}\|h\|_p, \tag{3.1}$$

$$\|f - f_N\|_p \le \frac{\vartheta^N C_p}{1 - \vartheta}\|h\|_p, \tag{3.2}$$

$$\|f - v\|_p \le \frac{C_p}{1 - \vartheta}\|Hv - m(x, v) - h\|_p. \tag{3.3}$$

Proof. The Nemytskii map $f \mapsto m(\cdot, f)$ is well defined on real L^p and has Lipschitz constant k. Its composition with H is a contraction with constant ϑ. The contraction theorem gives existence, uniqueness and (3.1). Summing the geometric tail of successive differences gives (3.2). For an arbitrary v, subtract its equation from the exact one, apply H, and use $H^2 = I$:

$$\|f - v\|_p \le \vartheta\|f - v\|_p + C_p\|Hv - m(x, v) - h\|_p.$$

This proves the residual estimate, without assuming that v is an iterate. □

3.2 The Cayley transformation and the zero mode

Put

$$\alpha_0 = \sqrt{\lambda\Lambda}, \quad C = A/\alpha_0, \quad s = \sqrt{\Lambda/\lambda}, \quad M = (I - C)(I + C)^{-1}, \quad k = \frac{s - 1}{s + 1}.$$

Symmetry and the spectral theorem give $\|M\|_{\mathrm{op}} \le k$, since the eigenvalues of C lie in $[1/s, s]$. Write $G = F/\alpha_0$. If $e = \nabla u$ has zero mean and $j = Ce - G$ is divergence free, define $f = e + j$. The Helmholtz constraints imply $Pf = e$, $Hf = e - j$, including at frequency zero: e has no zero mode, whereas j may have one. Eliminating e and j gives

$$Hf = Mf + h, \qquad h = 2(I + C)^{-1}G, \qquad e = Pf, \quad j = (I - P)f. \tag{3.4}$$

Conversely, (3.4) implies $j = Ce - G$, so this correspondence is exact in L^p. In particular no mean flux has been discarded.

For $s > 1$, $kC_p < 1$ is equivalent to both

$$p - 1 < \frac{s+1}{s-1}, \qquad \frac{1}{p-1} < \frac{s+1}{s-1},$$

which is precisely (1.2). If $s = 1$, $M = 0$ and every $1 < p < \infty$ is allowed. Since

$$\|h\|_p \le \frac{2}{\lambda(s+1)}\|F\|_p, \qquad \|P\|_p \le \frac{1+C_p}{2},$$

Theorem 3.1 proves (1.3). Poincare's inequality supplies the normalized periodic potential; the Euclidean closed L^p gradient is a homogeneous Sobolev gradient.

At $p = 2$ the bound is $1/\lambda$. For $F \in L^2(Q)$, the variational solution is the unique energy solution. If also $F \in L^p$ with $p < 2$, its gradient and flux lie in L^p, and uniqueness in (3.4) identifies it with the constructed solution. If $p > 2$, the constructed L^p solution lies in the energy space and the same conclusion follows. For $p < 2$ and arbitrary L^p data, approximate by $L^p \cap L^2$ data; the bound proves convergence to the bounded extension in $W^{1,p}$. This extension, rather than an unjustified energy identity, is the assertion for those data.

Example 3.2 (An interval with visible endpoints)**.** For $\lambda = 1, \Lambda = 9$, one has $s = 3$, $k = 1/2$. The permitted interval is $3/2 < p < 3$. At the upper endpoint the radial example below has gradient proportional to $r^{-2/3}$ and an infinite L^3 norm, although its energy is finite. For a constant matrix in the same coefficient class, much more regularity can hold. Sharpness is a uniform-class assertion.

3.3 Uniformly convex scalar nonlinear equations

Theorem 3.3 (Uniformly convex integrands)**.** *Suppose $W(x, z)$ is measurable in x, C^2 in z, $D_zW(x, 0) = 0$, and $\lambda I \le D_z^2 W(x, z) \le \Lambda I$. For every p in* (1.2) *and $F \in L^p(Q; \mathbb{R}^d)$ there is a unique mean-zero $u \in W^{1,p}(Q)$ satisfying*

$$\operatorname{div}(D_zW(x, \nabla u)) = \operatorname{div} F.$$

It obeys (1.3)*. The homogeneous Euclidean assertion holds as well. This is an analytic inversion theorem; it does not supply a finite implementation for an unrepresented measurable integrand.*

Proof. Set $C_x(z) = D_zW(x, z)/\alpha_0$. The map $z \mapsto z + C_x(z)$ is globally invertible: it is the gradient of $|z|^2/2 + W(x, z)/\alpha_0$, whose Hessian is bounded below by $(1 + 1/s)I$. Subtracting an arbitrary linear functional gives a coercive strictly convex function, with a unique minimizer. Its inverse T_x is C^1, with

$$DT_x(w) = (I + DC_x(T_x(w)))^{-1}.$$

Measurability follows, for example, by minimizing on successively finer rational grids in bounded balls and using uniqueness. The reflected inverse $\mathcal{R}_x(w) = 2T_x(w) - w$ has symmetric derivative with eigenvalues $(1 - c)/(1 + c)$, $c \in [1/s, s]$. Integrating this derivative on line segments gives the global Lipschitz constant k, and $\mathcal{R}_x(0) = 0$.

For $G = F/\alpha_0$, put

$$m(x, f) = \mathcal{R}_x(f + G) - \mathcal{R}_x(G), \qquad h = \mathcal{R}_x(G) + G = 2T_x(G).$$

The equation $Hf = m(x, f) + h$ is equivalent to $Hf = \mathcal{R}_x(f+G)+G$. Since $Hf = 2Pf - f$, this is equivalent to $Pf = T_x(f + G)$. Thus, with $e = Pf$ and $j = f - e$, $j = C_x(e) - G$; the exact gradient/divergence constraints prove the desired equation. Conversely every $W^{1,p}$ solution produces such an f, so uniqueness follows from Theorem 3.1. Finally $|T_x(G)| \le s|G|/(s + 1)$; the same estimates as in (3.4) prove the bound. □

3.4 Sharpness of the scalar interval

We use the classical radial singular solution and dimensional descent [14, Section 5], including the cutoff forcing needed for the global periodic and Euclidean estimates. Assume $s > 1$, and put $\nu = 1/s$. In a planar disk, let

$$A = \Lambda e_r \otimes e_r + \lambda e_\theta \otimes e_\theta, \qquad u_0 = r^\nu \cos\theta.$$

A polar-coordinate calculation gives $\operatorname{div}(A\nabla u_0) = (\Lambda\nu^2 - \lambda) r^{\nu-2} \cos\theta = 0$ off the origin. More explicitly, with $\psi = -\Lambda\nu r^\nu \sin\theta$,

$$A\nabla u_0 = \mathsf{J}\nabla\psi.$$

The potentials belong to $W^{1,2}$ near the origin, because $\int_0^1 r^{2\nu-1}\mathrm{d}r < \infty$. Their distributional derivatives therefore give the same equation across the origin, with no point source.

Choose a smooth cutoff η supported inside the disk and equal to one near zero. Set

$$u = \eta u_0, \qquad F = u_0 A\nabla\eta - \psi\mathsf{J}\nabla\eta. \tag{3.5}$$

The forcing is bounded and supported in an annulus, while $A\nabla u - F = \mathsf{J}\nabla(\eta\psi)$ is divergence free globally. Extend A, with the same ellipticity bounds, outside the disk. This constructs a periodic example by putting the disk inside Q, as well as a Euclidean example. Near the origin $|\nabla u|$ is comparable to $r^{\nu-1}$. Thus its L^p norm is infinite precisely when

$$p \geq \frac{2}{1-\nu} = \frac{2s}{s-1}.$$

The solution has finite energy, so a uniform strong upper-endpoint estimate is impossible. Its critical tail is compatible with the local weak endpoint proved below.

In Q of higher dimension use independence of the extra coordinates and a block-diagonal extension of A. In $\mathbb{R}^d$ multiply u and the first two components of F by an extra-coordinate cutoff. Complete F in the remaining components by the corresponding entries of $A\nabla u$. They are bounded, since u_0 is bounded; the remaining flux still has only planar divergence-free components. This gives the same local endpoint obstruction in every dimension.

For the lower endpoint, let $T_A : F \mapsto \nabla u$ be the energy solution operator on L^2. Symmetry of A gives, for $F, G \in L^2$,

$$\int_Q G \cdot T_A F = \int_Q (AT_A G) \cdot T_A F = \int_Q F \cdot T_A G.$$

If this operator had a uniform bounded extension on an exponent $p \leq 2s/(s+1)$, duality would give the same bound on $p' \geq 2s/(s-1)$, agreeing with the energy operator on the overlap. The preceding bounded-forcing example contradicts that bound. The same argument applies in Euclidean space. This completes Theorem 1.1.

3.5 The attained local weak endpoint

Failure of a strong estimate at $p_+ = 2s/(s-1)$ does not preclude control of the distribution function at that exponent. The radial solution above has exactly the critical power tail. To prove that every homogeneous energy solution has at most this tail, we first gain a strictly subcritical field moment. Its potentials then have a supercritical moment, which makes the errors caused by localization decay faster than the desired tail.

Only ordinary L^2 products enter this argument. The critical positive-product theorem of Section 4 is unnecessary here. Interior scalar and antisymmetric potentials are supplied by the

regularized Poincare operators [10, Corollary 3.4, Remark 3.5 and Proposition 4.1(i)]: on a fixed ball, if $\operatorname{curl} E = 0$, $\operatorname{div} B = 0$ and $E, B \in L^q$, $1 < q < \infty$, one can choose

$$\nabla v = E, \qquad B_i = \sum_j \partial_j \Psi_{ij}, \qquad \Psi_{ij} = -\Psi_{ji}, \qquad \|v\|_{W^{1,q}} + \|\Psi\|_{W^{1,q}} \le C(d,q)(\|E\|_q + \|B\|_q). \tag{3.6}$$

Fix the mean of v and the gauge of Ψ. The closed one-form and closed $(d-1)$-form interpretations give precisely these two operators on a ball; there is no cohomological term or boundary inverse.

Lemma 3.4 (Local subcritical bootstrap)**.** *Suppose C is measurable and symmetric on B_4, $s^{-1} I \le C \le sI$, $s > 1$, and*

$$E = \nabla u, \qquad B = CE, \qquad \operatorname{div} B = 0, \qquad u \in W^{1,2}(B_4).$$

For each fixed $2 < p < p_+$,

$$\|(|E|^2 + |B|^2)^{1/2}\|_{L^p(B_3)} \le C(d,s,p) \|(|E|^2 + |B|^2)^{1/2}\|_{L^2(B_4)}.$$

Proof. Suppose first that the two fields belong to L^q on an outer ball, with $q \ge 2$. Choose the potentials (3.6) there and a smooth cutoff χ supported in that ball and equal to one on a smaller concentric ball. Extend by zero the fields

$$a = \nabla(\chi v) = \chi E + g_a, \qquad b_i = \sum_j \partial_j(\chi \Psi_{ij}) = \chi B_i + (g_b)_i, \quad g_a = v \nabla \chi, \quad (g_b)_i = \sum_j \Psi_{ij} \partial_j \chi.$$

They are respectively curl free and divergence free, and are in $L^2(\mathbb{R}^d)$. Hence $f = a + b$ satisfies $Hf = a - b$. Extend $M = (I - C)(I + C)^{-1}$ by zero outside the outer ball. Since $(I - C) = M(I + C)$,

$$Hf - Mf = (I - M) g_a - (I + M) g_b =: h, \qquad \|M\|_\infty \le \frac{s-1}{s+1}. \tag{3.7}$$

Sobolev embedding gives $h \in L^a \cap L^2$ whenever $2 \le a < p_+$ and $a \le dq/(d-q)$ if $q < d$; every finite $a < p_+$ is available if $q \ge d$. At $q = d$ only finite exponents are used.

Theorem 3.1 inverts (3.7) in both L^2 and L^a. These inverses agree on the overlap: their identical Neumann partial sums converge in both spaces, and therefore to the same distribution. Thus the existing f belongs to L^a, with a bound by the outer L^q norms. On $\chi = 1$, the identities $E = (f + Hf)/2$ and $B = (f - Hf)/2$ give the improved field bound.

Starting at $q = 2$, use finitely many nested balls between B_4 and B_3. While $q < d$, reduce $1/q$ at each step by $1/(2d)$, or by the smaller amount that reaches the prescribed $1/p$. Such a step stays strictly within the Sobolev range. Once $q \ge d$, reach p in the next step. Stop upon reaching p, so every inversion exponent remains strictly below p_+. This takes finitely many steps depending only on d, p. Choose equally spaced intermediate radii for that number of steps. The potential, cutoff and inverse bounds then give the stated constant. No bound uniform as $p \uparrow p_+$ is asserted. □

Lemma 3.5 (Exponents for the cutoff errors)**.** *For $p_+ > 2$, put*

$$a = p_+ + \frac{p_+^2}{2(d + p_+)}, \qquad p = \frac{1}{2}\left[p_+ + \max\left\{2, \frac{da}{d+a}, \frac{a(p_+ - 1)}{a-1}, \frac{a(p_+ - 2)}{a-2}\right\}\right].$$

Then $2 < p < p_+ < a < \infty$, $W^{1,p}(B_3) \hookrightarrow L^a(B_3)$, and

$$\theta_1 = p(1 - 1/a) - 1 > p_+ - 2, \qquad \theta_2 = p(1 - 2/a) > p_+ - 2. \tag{3.8}$$

Proof. All four entries in the maximum are strictly below p_+. For $da/(d+a)$ this is immediate if $d \le p_+$. If $d > p_+$, substitute the formula for a into $a(d-p_+) < dp_+$. The two remaining inequalities follow from $a > p_+$ by clearing their positive denominators. The strict inequality $p > da/(d+a)$ gives the asserted Sobolev embedding, including the cases $p \ge d$. The last two entries in the maximum are exactly the conditions in (3.8). □

Proof of Theorem 1.2. Translate and dilate to B_4, using $\widetilde{u}(y) = r^{-1}u(x_0 + ry)$, and suppress the tilde. Set

$$C = A/\sqrt{\lambda\Lambda}, \qquad E = \nabla u, \qquad B = CE,$$
$$V = (|E|^2 + |B|^2)^{1/2}, \qquad J = E \cdot B, \qquad R = (|E+B| + |E-B|)/2.$$

The eigenvalues of C lie in $[1/s, s]$. The identity $R^2/J + J/R^2 = V^2/J$, or its direct algebraic interpretation when a field vanishes, and $1 + c^2 \le (s + s^{-1})c$ for these eigenvalues show that

$$R^2 \le sJ, \qquad |E|, |B| \le R \le V \le \sqrt{1+s^2}\,|E|. \tag{3.9}$$

For completeness, if $\alpha = |E+B|/2$ and $\gamma = |E-B|/2$, then $J = \alpha^2 - \gamma^2 \ge 0$ and $R = \alpha + \gamma$. Thus $R^2/J \ge 1$, and monotonicity of $x + x^{-1}$ on $[1, \infty)$ proves the implication. If $J = 0$, ellipticity forces both fields to vanish.

If $\|V\|_{L^2(B_4)} = 0$, there is nothing to prove. Otherwise divide both fields by this norm. Lemmas 3.4–3.5 give $\|V\|_{L^p(B_3)} \le C(d, s)$. Choose potentials on B_3, a cutoff $0 \le \chi \le 1$ equal to one on B_2, and the compact fields a, b as in (3.7). With

$$r = \frac{|a+b| + |a-b|}{2}, \qquad D = a \cdot b, \qquad G = (|g_a|^2 + |g_b|^2)^{1/2},$$

we have

$$\|r\|_p + \|G\|_a \le C(d, s), \qquad a, b \in L^2, \qquad \int_{\mathbb{R}^d} D \,\mathrm{d}x = 0. \tag{3.10}$$

Indeed the potentials have their L^a bounds by Sobolev embedding. The zero product integral follows by testing $\operatorname{div} b = 0$ against the compactly supported $W^{1,2}$ potential χv, using ordinary L^2 approximation.

We retain the precise distortion coefficient under localization. The map $(a, b) \mapsto r(a, b)$ is a norm and $r(g_a, g_b) \le G$. Consequently $\chi R \le r + G$ and

$$r^2 \le s\chi^2 J + 2rG + 3G^2.$$

To see the last bound directly, use $r \le \chi R + G$, then $\chi R \le r + G$ in the cross term. Moreover

$$D = \chi^2 J + C_0, \qquad C_0 = \chi E \cdot g_b + \chi B \cdot g_a + g_a \cdot g_b.$$

Since $\chi V \le \sqrt{2}\,\chi R \le \sqrt{2}(r + G)$,

$$|C_0| \le \sqrt{2}\, rG + (\sqrt{2} + 1/2)G^2.$$

It follows that

$$D \ge \frac{1}{s} r^2 - c_s(rG + G^2). \tag{3.11}$$

Replacing $1/s$ by a smaller coefficient would lose the endpoint.

Define the finite C^1 function

$$F(t) = \int_{\mathbb{R}^d} (r - t)_+^2 \,\mathrm{d}x, \qquad F'(t) = -2 \int_{\mathbb{R}^d} (r - t)_+ \,\mathrm{d}x, \qquad t > 0.$$

The compact calibration, Corollary 2.4, and (3.10) imply

$$\int_{\{r>t\}} (r-t)(D/r-t)\,\mathrm{d}x = \int_{\mathbb{R}^d} (D - L_t(a,b))\,\mathrm{d}x \le 0.$$

All terms are integrable by $a, b \in L^2$. Insert (3.11) and use $(r-t)/r \le 1$. Since

$$\int_{\{r>t\}} (r-t)(r/s-t)\,\mathrm{d}x = \frac{1}{s}\left(F(t) + \frac{s-1}{2} tF'(t)\right),$$

we obtain

$$F(t) + \frac{s-1}{2} tF'(t) \le C_s \int_{\{r>t\}} (rG + G^2)\,\mathrm{d}x. \tag{3.12}$$

For $t \ge 1$, Holder and (3.10) give

$$\int_{\{r>t\}} rG \le \|G\|_a \left(\int_{\{r>t\}} r^{a/(a-1)}\right)^{(a-1)/a} \le Ct^{-\theta_1},$$
$$\int_{\{r>t\}} G^2 \le \|G\|_a^2 |\{r>t\}|^{1-2/a} \le Ct^{-\theta_2}.$$

The first estimate uses $p > a/(a-1)$; in both estimates $\int r^p \le C$. Put $\kappa = p_+ - 2 = 2/(s-1)$ and $\eta = \min\{\theta_1, \theta_2\} - \kappa > 0$. Equation (3.12) becomes

$$tF'(t) + \kappa F(t) \le Ct^{-\kappa-\eta}, \qquad (t^\kappa F(t))' \le Ct^{-1-\eta}.$$

Integrating from 1 to t, and using $F(1) \le \|r\|_2^2 \le C$, proves

$$F(t) \le Ct^{-\kappa}, \qquad |\{r > 2t\}| \le t^{-2}F(t) \le Ct^{-p_+} \quad (t \ge 1).$$

On B_2, $r = R$, so (3.9) proves the weak bounds for both fields on B_1. For thresholds below 2 the volume of B_1 supplies the bound. Restore the field norm, use $\|V\|_2 \le \sqrt{1+s^2}\|E\|_2$, and restore the spatial dilation. Under the chosen rescaling the gradient and normalized flux have unchanged values at corresponding points, whereas both a level-set measure and its ball volume acquire the factor r^d. Likewise $⨍_{B_4} |\nabla \widetilde{u}|^2 = ⨍_{B_{4r}(x_0)} |\nabla u|^2$. The fixed volumes of B_1, B_4 therefore give exactly (1.4). □

Sharpness within the homogeneous class

Use the uncut solution $u_0 = r^{1/s}\cos\theta$ from Section 3.4, on the entire planar B_4. It solves the homogeneous equation, including at zero, and has finite energy. Since

$$s^{-1}r^{1/s-1} \le |\nabla u_0| \le r^{1/s-1},$$

its distribution on B_1, for all sufficiently large t, satisfies

$$c_s t^{-p_+} \le |\{x \in B_1 : |\nabla u_0| > t\}| \le C_s t^{-p_+}.$$

The normalized flux has the same comparison, by ellipticity. In dimension $d > 2$, take independence of the remaining coordinates and use the block λI there. The two-dimensional singularity remains locally square integrable. An inner cylinder in B_1 and the containing cylinder give the same two tail bounds by Fubini. These are homogeneous solutions on B_4; the forcing cutoff used for the global strong obstruction is absent.

Thus no weak exponent larger than p_+ holds uniformly. Nor can the endpoint be replaced by $L^{p_+,b}$ for any finite Lorentz index $0 < b < \infty$. In the distribution-function convention, finiteness of this Lorentz quasinorm requires

$$\int_0^\infty \left[t\,|\{|g| > t\} \cap B_1|^{1/p_+} \right]^b \frac{\mathrm{d}t}{t} < \infty.$$

The lower tail bound makes its integral at infinity diverge. In particular the strong L^{p_+} endpoint fails, consistently with the maximal open interval.

Corollary 3.6 (Homogeneous uniformly convex equations)**.** *Let W satisfy the hypotheses of Theorem 3.3, with $s > 1$. If $u \in W^{1,2}(B_{4r})$ satisfies* $\operatorname{div} D_z W(x, \nabla u) = 0$*, then* (1.4) *holds for both ∇u and $D_z W(x, \nabla u)/\sqrt{\lambda\Lambda}$, with the same right-hand side and dependence $C(d, s)$.*

Proof. The measurable symmetric matrix

$$A_u(x) = \int_0^1 D_z^2 W(x, t\nabla u(x))\,\mathrm{d}t$$

has the bounds $\lambda I \le A_u \le \Lambda I$. The normalization $D_z W(x, 0) = 0$ gives $D_z W(x, \nabla u) = A_u \nabla u$. Apply Theorem 1.2 to this existing energy solution. The linear examples above are already uniformly convex quadratic integrands, so they also test the sharpness of this class. □

3.6 A represented finite Fourier solver

For $m(x, z) = M(x)z$, suppose M, h are rational trigonometric polynomials, real as fields, with a certified $\|M\|_\infty \le k$ and $\vartheta = kC_p < 1$. Finite convolution and the rational symbol (1.1) compute every Picard iterate exactly. If H_0 is a rational upper bound for $\|h\|_p$, then the gradient error is bounded by

$$\|Pf - Pf_N\|_p \le \frac{(1 + C_p)C_p}{2(1 - \vartheta)} \vartheta^N H_0. \tag{3.13}$$

Choose N from this bound before starting. An optional residual test uses (3.3), multiplied by $(1 + C_p)/2$, and a rational Fourier-coefficient bound for the residual norm. Such a coefficient bound need not itself tend to zero; it must not replace the a priori iteration cap.

The concrete example reconstructed in Appendix A has $d = 2, p = 4, k = 3/10$,

$$M = \frac{3}{10} \begin{pmatrix} \cos(2\pi x_1) & \sin(2\pi x_1) \\ \sin(2\pi x_1) & -\cos(2\pi x_1) \end{pmatrix}, \qquad h = \begin{pmatrix} \cos(2\pi x_2) \\ \sin(2\pi(x_1 + x_2)) \end{pmatrix}.$$

Exact convolution gives $M^2 = k^2 I$, hence $\|M\|_\infty = k$. Its coefficient and forcing are

$$A = \frac{(1 + k^2)I - 2M}{1 - k^2}, \qquad F = \frac{(I - M)h}{1 - k^2}.$$

They satisfy $A(I + M) = I - M$ and $2(I + A)^{-1}F = h$ exactly. The ellipticity endpoints are $7/13, 13/7$; the scalar interval is $13/10 < p < 13/3$. The accompanying script verifies the differential constraints and an exact rational L^4 error certificate.

4 Critical products, localization and logarithmic regularity

4.1 The distortion geometry and the initial integrability

For any pair with $J \ge 0$, write $\alpha = |E + B|/2, \gamma = |E - B|/2$. Then

$$J = \alpha^2 - \gamma^2, \quad R = \alpha + \gamma, \quad V^2 = 2(\alpha^2 + \gamma^2), \qquad J \le R^2 \le V^2 \le 2R^2.$$

If $J > 0$, put $\kappa = R^2/J \geq 1$. The identity $V^2/J = \kappa + \kappa^{-1}$, and monotonicity of $x + x^{-1}$ on $[1, \infty)$, prove

$$R^2 \leq KJ \quad \Longleftrightarrow \quad V^2 \leq (K + K^{-1})J.$$

For $J = 0$, either inequality forces both fields to vanish.

On a finite-volume subdomain D, Holder's inequality gives, for $1 < q < 2$ and $m = q/(2-q)$,

$$\left(\fint_D V^q \, \mathrm{d}x \right)^{2/q} \leq 2\langle J \rangle_D \left(\langle K^m \rangle_D \right)^{1/m} \leq \frac{2m}{\mathrm{e}\tau} \langle J \rangle_D \left(\langle \mathrm{e}^{\tau K} \rangle_D \right)^{1/m}. \tag{4.1}$$

The last step uses $t^m \leq (m/(\mathrm{e}\tau))^m \mathrm{e}^{\tau t}$. This proves every subquadratic local field moment before any weak integration by parts is used. There is also the critical estimate

$$\frac{R^2}{\log(\mathrm{e} + R)} \leq \frac{2}{\tau} J + \mathrm{e}^{\tau K}. \tag{4.2}$$

Indeed, on $K \leq (2/\tau) \log(\mathrm{e} + R)$ use distortion. On the complement, $\mathrm{e}^{\tau K} > (\mathrm{e} + R)^2$. Thus E and B belong locally to the Zygmund class denoted here by $L^2 \log^{-1} L$:

$$F \in L^2 \log^{-1} L(D) \quad \text{means} \quad \int_D \frac{|F|^2}{\log(\mathrm{e} + |F|)} \, \mathrm{d}x < \infty.$$

Changing the positive constant inside the logarithm gives the same class.

4.2 The imported critical-product theorem

We state the precise working version of the critical positive-product result used below. It is the $p = 2$ nonnegative case of Sbordone [17, Corollary 3.3 and the following discussion, p. 745]. Positivity is part of the import; a formal div–curl identity for arbitrary subquadratic fields is not being assumed.

Theorem 4.1 (Positive critical div–curl product)**.** *Let $U, W \in L^2 \log^{-1} L_{\mathrm{loc}}(D; \mathbb{R}^d)$, with $\operatorname{curl} U = 0$, $\operatorname{div} W = 0$. Suppose their pointwise scalar product is nonnegative and locally integrable. If u is a local scalar potential of U, the canonical distributional product is the pointwise product:*

$$\operatorname{div}(uW) = U \cdot W. \tag{4.3}$$

Here one may take $u \in W^{1,q}_{\mathrm{loc}}$ for any $q < 2$ sufficiently close to two, so that $uW \in L^1_{\mathrm{loc}}$; additive constants do not affect the distribution.

All its hypotheses hold for the pair in Theorem 1.3: (4.2) gives the two critical classes, and $J \geq 0$ and $J \in L^1_{\mathrm{loc}}$ are explicit assumptions. For $q = 2 - 1/(d+1)$, the Sobolev exponent $q^* = dq/(d-q)$ satisfies

$$q^* > 2, \qquad \frac{1}{q} + \frac{1}{q^*} < 1.$$

Thus the canonical expression is concretely an L^1 vector field inside a divergence. This also specifies the convention in (4.3); it is not an unspecified limiting product.

4.3 Constraint-preserving localization

By a dilation and by dividing both fields by A_0, it is enough to work on B_2, with

$$\langle \mathrm{e}^{\tau K} \rangle_{B_2} = M_0, \qquad (\langle V^q \rangle_{B_2})^{1/q} \leq 1.$$

This follows exactly from (4.1) with $m = 2d + 1$. The zero-energy case was already settled by distortion.

We use the following standard consequence of regularized Poincare operators [10, Corollary 3.4, Remark 3.5 and Proposition 4.1(i)]. On a ball, a curl-free L^q vector field has a scalar potential u, and a divergence-free L^q vector field has an antisymmetric potential Ψ, with

$$\begin{aligned} &\nabla u = E, \qquad B_i = \sum_j \partial_j \Psi_{ij}, \qquad \Psi_{ij} = -\Psi_{ji}, \\ &\|u\|_{W^{1,q}(B_2)} + \|\Psi\|_{W^{1,q}(B_2)} \le C(d,q)\|V\|_{L^q(B_2)}. \end{aligned} \tag{4.4}$$

For u, fix a zero mean. For Ψ, the regularized operator fixes a gauge. To match the differential-form statement, apply the Poincare operator to the closed one-form $E \cdot \mathrm{d}x$, and to the closed $(d-1)$-form obtained from B by interior multiplication with the volume form. A ball is starlike, so there is no cohomological term. The operators have order -1, giving $L^q \to W^{1,q}$. Only interior potentials are used; no boundary trace or compact-support inverse is imported.

Let $0 \le \chi \le 1$ be smooth, supported in B_2, and equal to one on B_1. Extend the following fields by zero:

$$a = \nabla(\chi u) = \chi E + g_a, \qquad b_i = \sum_j \partial_j(\chi \Psi_{ij}) = \chi B_i + (g_b)_i, \quad g_a = u\nabla\chi, \quad (g_b)_i = \sum_j \Psi_{ij}\partial_j\chi.$$

They are compactly supported and exactly satisfy $\operatorname{curl} a = 0, \operatorname{div} b = 0$. Set $G^2 = |g_a|^2 + |g_b|^2$. By (4.4) and Sobolev embedding,

$$\|g_a\|_{q^*} + \|g_b\|_{q^*} \le C(d).$$

For some $\delta = \delta(d) > 0$, the identities

$$D = a \cdot b = J_0 + C_0, \qquad J_0 = \chi^2 J, \qquad C_0 = \chi E \cdot g_b + \chi B \cdot g_a + g_a \cdot g_b \tag{4.5}$$

therefore give

$$\|C_0\|_{1+\delta} + \|G^2\|_{1+\delta} \le C(d), \qquad \|r\|_q \le C(d), \qquad r = r(a,b). \tag{4.6}$$

For instance take $1+\delta$ no larger than $\min\{(1/q + 1/q^*)^{-1}, q^*/2\}$, which is strictly greater than one.

The product in (4.5) is also the canonical product. To check this without assuming L^2 fields, use (4.3) for uB, multiply by χ^2, and apply the Sobolev product rule to the additional terms involving Ψ and χ. Those derivatives have the integrability just verified: $u, \Psi, g_a, g_b \in L^{q^*}$ and $E, B, \nabla\Psi \in L^q$, with $1/q + 1/q^* < 1$. More explicitly, antisymmetry gives $\operatorname{div} g_b = -B \cdot \nabla\chi$. Thus

$$\operatorname{div}((\chi u)b) = \chi^2 J + 2\chi u B \cdot \nabla\chi + (\chi E + u\nabla\chi) \cdot g_b - \chi u B \cdot \nabla\chi = J_0 + C_0.$$

It has compact support and an L^1 right-hand side. Testing with a smooth function equal to one on that support proves

$$\int_{\mathbb{R}^d} D \,\mathrm{d}x = 0. \tag{4.7}$$

For every $\eta > 0$, the triangle inequality for the norm $r(a,b)$ and the scalar inequality $(x+y)^2 \le (1+\eta)x^2 + (1+\eta^{-1})y^2$ give

$$r^2 \le (1+\eta)KJ_0 + C_\eta G^2. \tag{4.8}$$

In addition $r \in L^s$ for every $s < 2$, by (4.1), with the potential bounds at each such exponent and a fixed choice of potentials bounded in these spaces. The regularized operators can be chosen once and have these bounds simultaneously. This last observation is used only to justify the initial damped integrals.

4.4 Damping and the two sharp coefficients

Fix $\beta > 0$, $0 < \varepsilon < 1$, and use the positive weight

$$w_\varepsilon(t) = t^{-1-\varepsilon}(\log t)^{\beta-1}, \qquad t \geq \mathrm{e}.$$

Define, as zero when $r < \mathrm{e}$,

$$A_\varepsilon(r) = \int_{\mathrm{e}}^{r} (1 - t/r) w_\varepsilon(t) \mathrm{d}t, \tag{4.9}$$

$$B_\varepsilon(r) = \int_{\mathrm{e}}^{r} t(r - t) w_\varepsilon(t) \mathrm{d}t. \tag{4.10}$$

For $t < r$, a direct rearrangement of (2.3) gives

$$D - L_t = (r - t)(D/r - t).$$

Consequently the threshold integral is

$$\int_{\mathrm{e}}^{\infty} (D - L_t) w_\varepsilon(t) \mathrm{d}t = D A_\varepsilon(r) - B_\varepsilon(r). \tag{4.11}$$

This is an absolutely integrable space–threshold expression. Indeed A_ε is bounded for fixed $\varepsilon > 0$, and

$$B_\varepsilon(r) \leq C_{\beta,\varepsilon}(1 + r^{2-\varepsilon/2}).$$

The latter bound follows by absorbing a logarithmic power into $t^{\varepsilon/2}$. The initial subquadratic moments justify it in space. Thus Fubini is valid before proving any improved moment. Corollary 2.4 and (4.7) imply

$$\int D A_\varepsilon(r) \, \mathrm{d}x \leq \int B_\varepsilon(r) \, \mathrm{d}x. \tag{4.12}$$

Lemma 4.2 (Uniform large-r threshold estimates)**.** *Given $\beta > 0$ and $0 < \eta \leq 1/8$, there are $0 < \varepsilon_0 \leq \eta/16$ and $r_0 > \mathrm{e}$, depending only on β, η, such that for $0 < \varepsilon \leq \varepsilon_0$, $r \geq r_0$,*

$$A_\varepsilon(r) \geq \frac{1 - \eta}{\beta} r^{-\varepsilon} (\log r)^\beta, \tag{4.13}$$

$$B_\varepsilon(r) \leq \frac{1 + \eta}{2} r^{2-\varepsilon} (\log r)^{\beta-1}. \tag{4.14}$$

For all $r \geq \mathrm{e}$, $A_\varepsilon(r) \leq (\log r)^\beta/\beta$.

Proof. In (4.9), $t^{-\varepsilon} \geq r^{-\varepsilon}$. Hence

$$A_\varepsilon(r) \geq r^{-\varepsilon} \left[\frac{(\log r)^\beta - 1}{\beta} - \frac{1}{r} \int_{\mathrm{e}}^{r} (\log t)^{\beta-1} \mathrm{d}t \right].$$

The second term is $o((\log r)^\beta)$, independently of ε. For $\beta \geq 1$ it is at most $(\log r)^{\beta-1}$; for $0 < \beta < 1$, split at $\sqrt{r}$ to bound it by $r^{-1/2} + 2(\log r)^{\beta-1}$. Choosing r_0 large proves (4.13). The upper bound follows by dropping $1 - t/r$ and $t^{-\varepsilon}$.

For B_ε, substitute $t = rv$. If $\beta \geq 1$,

$$B_\varepsilon(r) \leq r^{2-\varepsilon} (\log r)^{\beta-1} \int_0^1 (1 - v) v^{-\varepsilon} \mathrm{d}v, \qquad \int_0^1 (1 - v) v^{-\varepsilon} \mathrm{d}v = \frac{1}{(1 - \varepsilon)(2 - \varepsilon)}.$$

Choose ε_0 small enough to make the last factor at most $(1 + \eta)/2$. For $0 < \beta < 1$, choose $0 < \delta_0 < 1/2$ such that $(1 - \delta_0)^{\beta-1} \leq 1 + \eta/4$. On $t \geq r^{1-\delta_0}$ the same integral is multiplied

by at most $1+\eta/4$. Choose $\varepsilon_0 \le 1/4$ so that its factor is at most $(1+\eta/4)/2$. The omitted interval contributes, relative to $r^{2-\varepsilon}(\log r)^{\beta-1}$, at most

$$\frac{r^{-\delta_0(1-\varepsilon)}}{1-\varepsilon}(\log r)^{1-\beta},$$

which tends uniformly to zero for the chosen ε. Increase r_0 until it is at most $\eta/8$. Since $(1+\eta/4)^2/2+\eta/8 \le (1+\eta)/2$ for $0<\eta\le 1/8$, this proves (4.14). Every threshold choice can be made using the displayed elementary inequalities; no limit is taken in ε here. □

4.5 Absorption exactly below the exponential threshold

Now fix $0<\beta<\tau$, and choose

$$\sigma=\frac{\beta+\tau}{2},\qquad \eta=\frac{\sigma-\beta}{8\sigma},\qquad a_*=\frac{1-\eta}{\beta},\qquad b_*=\frac{1+\eta}{2}.$$

These choices satisfy

$$c_* := a_* - \frac{2b_*(1+\eta)}{\sigma} \ge \frac{\sigma-\beta}{2\beta\sigma} > 0. \tag{4.15}$$

Indeed, with $x=\beta/\sigma\in(0,1)$, the numerator after multiplication by β is $1-\eta-x(1+\eta)^2 \ge 1-x-3\eta-\eta^2 \ge (1-x)/2$, because $\eta=(1-x)/8$.

Let

$$X_\varepsilon=\int_{\{r\ge r_0\}} J_0 r^{-\varepsilon}(\log r)^\beta \,\mathrm{d}x,\qquad Y_\varepsilon=\int_{\{r\ge r_0\}} r^{2-\varepsilon}(\log r)^{\beta-1}\,\mathrm{d}x.$$

Equations (4.5) and (4.12), followed by Lemma 4.2, give

$$a_*X_\varepsilon \le b_*Y_\varepsilon + C(d,\beta,\tau). \tag{4.16}$$

Here is the uniform error accounting. Since $D\le r^2$ and $J_0=D-C_0\ge 0$, the part with $r<r_0$ is bounded by a constant and $\int|C_0|$. Also

$$\int |C_0|A_\varepsilon(r) \le \frac{1}{\beta}\int |C_0|\log^\beta(\mathrm{e}+r) \le C(d,\beta),$$

using (4.6), Holder, and $\log^a(\mathrm{e}+r)\le C_{a,b}(1+r^b)$ for any $a,b>0$. The small-r contribution of B_ε is uniformly bounded by its undamped integral on a finite threshold interval. The same argument controls every integral of $G^2\log^a(\mathrm{e}+r)$, uniformly in ε.

On the set $K\le 2\log r/\sigma$, use (4.8) to obtain

$$r^{2-\varepsilon}(\log r)^{\beta-1} \le \frac{2(1+\eta)}{\sigma}J_0 r^{-\varepsilon}(\log r)^\beta + C_\eta G^2 r^{-\varepsilon}(\log r)^{\beta-1}.$$

No extra power of the distortion remains. On the complement $r^2<\mathrm{e}^{\sigma K}$; any positive logarithmic power is bounded by a polynomial in K, which is absorbed into $\mathrm{e}^{(\tau-\sigma)K}$. Thus

$$Y_\varepsilon \le \frac{2(1+\eta)}{\sigma}X_\varepsilon + C(d,\beta,\tau)(1+M_0). \tag{4.17}$$

The factor $r^{-\varepsilon}\le 1$ creates no new error since $r\ge r_0>\mathrm{e}$. Combine (4.16)–(4.17) and (4.15) to bound both X_ε and Y_ε by $C(d,\beta,\tau)(1+M_0)$, independently of $0<\varepsilon\le\varepsilon_0$.

As $\varepsilon\downarrow 0$, their nonnegative integrands increase. Monotone convergence therefore gives the undamped estimates. On B_1, $\chi=1$, $g_a=g_b=0$, so $r=R$ and $J_0=J$. The inequalities $J\le R^2$, $R^2\le V^2\le 2R^2$ compare the logarithms above and below, including when $\beta-1<0$. The bounded region $R<r_0$ contributes a constant. This proves the normalized estimate (1.7) after undoing the two normalizations. Covering compact subsets proves (1.6). No undamped Fubini step was used.

Corollary 4.3 (Energy concentration modulus)**.** *With the normalization of Theorem 1.3, for a measurable $S \subset B_r$, $v = |S|/|B_r| \in (0,1]$, and $0 < \beta < \tau$,*

$$\frac{1}{|B_r|}\int_S \frac{J}{A_0^2}\,\mathrm{d}x \le \frac{C(d,\beta,\tau)(1+M_0)}{\log^\beta(\mathrm{e}+1/v)}.$$

In particular these normalized energies are uniformly integrable when M_0 is bounded. If $1 < \beta < \tau$, the same conclusion holds for V^2/A_0^2, with logarithmic exponent $\beta - 1$.

Proof. Split J/A_0^2 at $v^{-1/2}$. The lower part contributes at most $\sqrt{v}$; bound the upper part by (1.7) divided by $\log^\beta(\mathrm{e}+v^{-1/2})$. These two terms are bounded by the stated expression, since $\sqrt{v}\log^\beta(\mathrm{e}+1/v)$ is bounded on $(0,1]$. For V^2/A_0^2, use its logarithmic estimate in the same way, observing that $\log(\mathrm{e}+V/A_0)$ is comparable to $\log(\mathrm{e}+V^2/A_0^2)$. The positive exponent requires $\beta > 1$. □

5 Endpoint examples and scalar consequences

The examples use a single radial construction, which also explains the threshold. A slowly vanishing radial profile makes the energy finite. Its logarithmic derivative determines the distortion. Choosing that derivative at the critical scale leaves precisely the divergent integral $\int^\infty (t\log t)^{-1}\mathrm{d}t$. This radial method has planar finite-distortion precedents [4]; it is used here to test both members of (1.6).

5.1 A reusable radial calculation

Work in $0 < r < \mathrm{e}^{-T} \subset \mathbb{R}^2$, let $t = \log(1/r)$, and choose a positive profile $\rho(r)$ tending to zero at the origin, with $\rho'(r) = h(t)\rho(r)/r$, $0 < h \le 1$. Set

$$u = \rho(r)\cos\theta, \quad v = \rho(r)\sin\theta, \quad E = \nabla u, \quad B = -\mathsf{J}\nabla v, \quad A = h^{-1}e_r \otimes e_r + h e_\theta \otimes e_\theta. \tag{5.1}$$

In the polar frame,

$$E = \frac{\rho}{r}(h\cos\theta, -\sin\theta), \qquad B = \frac{\rho}{r}(\cos\theta, -h\sin\theta) = AE.$$

Consequently, with $K = h^{-1}$,

$$J = \frac{h\rho^2}{r^2}, \quad V^2 = \frac{(1+h^2)\rho^2}{r^2}, \quad R = \frac{\rho}{r}, \quad R^2 = KJ. \tag{5.2}$$

Whenever $\rho \to 0$ and the displayed gradients are locally integrable, the definitions extend distributionally across the origin. Indeed the boundary term in the weak derivative of either continuous potential is bounded by $Cr\rho(r)$, which tends to zero. Mixed distributional derivatives then prove both constraints, including at the origin. No divergence-free assertion is inferred only from a punctured-domain calculation.

5.2 Failure at the logarithmic endpoint

Fix $\tau > 0$, and take T large enough that

$$\rho = t^{-\tau/2}(\log t)^{-1/2}, \qquad h = \frac{\tau + 1/\log t}{2t} < 1.$$

The derivative relation and all identities in (5.2) follow directly. The potentials are continuous at zero, their gradients are in L^1, and the finite energy is

$$\int_{r<\mathrm{e}^{-T}} J\,\mathrm{d}x = 2\pi\int_T^\infty h\rho^2\mathrm{d}t = \pi\frac{T^{-\tau}}{\log T}. \tag{5.3}$$

Here $(\rho^2)'(t) = -2h\rho^2$. The exponential moment is

$$\int_{r<\mathrm{e}^{-T}} \mathrm{e}^{\tau K}\,\mathrm{d}x = 2\pi \int_T^\infty \exp\left(-\frac{2t}{\tau\log t+1}\right)\mathrm{d}t < \infty. \tag{5.4}$$

For an explicit majorant, if $T \geq \mathrm{e}$, then $\tau\log t + 1 \leq (\tau+1)\sqrt{t}$. With $c = 2/(\tau+1)$, the last integral is at most

$$4\pi \mathrm{e}^{-c\sqrt{T}}\left(\frac{\sqrt{T}}{c} + \frac{1}{c^2}\right).$$

Since $\log(\mathrm{e}+J) \sim 2t$ and $\log(\mathrm{e}+V) \sim t$, the two endpoint integrals at $\beta = \tau$ are each comparable, in their tails, to

$$\int_T^\infty \frac{\mathrm{d}t}{t\log t} = \infty.$$

For the first, $h \sim \tau/(2t)$; for the second, $(1+h^2)\rho^2 t^{\tau-1} \sim (t\log t)^{-1}$. This proves simultaneous failure of both endpoint assertions.

In particular, for $\tau = 1$, finite energy and $\mathrm{e}^K \in L^1$ do not imply separate L^2 integrability. The angular integrals satisfy

$$\int_0^{2\pi} |E|^2\,\mathrm{d}\theta = \int_0^{2\pi} |B|^2\,\mathrm{d}\theta = \pi(1+h^2)\rho^2/r^2.$$

Their radial integrals contain $\int_T^\infty (t\log t)^{-1}\mathrm{d}t$. Thus this is a failure for each field, not merely their sum.

5.3 Every exponential moment without any power gain

Instead choose

$$\rho = \exp(-2\sqrt{t}), \qquad h = t^{-1/2}, \qquad K = \sqrt{t}, \qquad T > 1.$$

The same construction gives continuous potentials, valid global distributional constraints on the disk, and energy $\pi\exp(-4\sqrt{T})$. For every finite $\sigma > 0$,

$$\int \mathrm{e}^{\sigma K}\,\mathrm{d}x = 2\pi\int_T^\infty \exp(-2t+\sigma\sqrt{t})\mathrm{d}t < \infty.$$

On angular sectors of fixed positive size, either separate field has size at least $\rho/(2r)$. For $p > 2$, its p-moment therefore dominates a constant times

$$\int_T^\infty \exp((p-2)t - 2p\sqrt{t})\mathrm{d}t = \infty.$$

Thus even all finite exponential moments do not imply a general $L^{2+\delta}$ conclusion.

Both examples lift to any $d > 2$ by independence of the extra coordinates on a cylinder and by taking $A = \mathrm{diag}(A_{\mathrm{planar}}, I)$. The extra field components vanish. Fubini preserves every finite moment and every divergence just proved. The singular set becomes a codimension-two axis; the scalar potentials remain continuous there.

5.4 Consequences for existing scalar solutions

Let A be symmetric positive definite and satisfy, in fixed units, $K^{-1}I \leq A \leq KI$. For an existing weak solution u, put $E = \nabla u$, $B = AE$. If $J = E\cdot AE$ is locally integrable and $\mathrm{e}^{\tau K}$ is locally integrable, Theorem 1.3 applies. Indeed every eigenvalue $a \in [K^{-1}, K]$ satisfies

$$1 + a^2 \leq (K + K^{-1})a,$$

and summing in an orthonormal eigenbasis proves the distortion inequality. If $\tau > 1$, choose $1 < \beta < \tau$ to recover local L^2 integrability of both fields. The preceding endpoint shows why $\tau = 1$ alone is insufficient.

More generally the same conclusion holds for an existing solution of $\operatorname{div} \mathcal{A}(x, \nabla u) = 0$ whenever its actual pair $(\nabla u, \mathcal{A}(x, \nabla u))$ has the stated integrable nonnegative product and distortion. No general degenerate nonlinear existence or uniqueness theorem follows from this observation. A pointwise condition number alone is also insufficient for the stated hypothesis: it does not control the absolute scale of A and its inverse. Multiplying A by an arbitrary function of x changes the divergence equation.

6 Degenerate periodic response and exact error identities

Assume (1.8) and put $\mu = M/\tau$. The elementary bounds

$$\int_Q K \,\mathrm{d}x \le \mu, \qquad \int_Q K^r \,\mathrm{d}x \le r! M/\tau^r \quad (r = 1, 2, \ldots) \tag{6.1}$$

will suffice for all subquadratic embeddings. In particular $\mu \ge 1$. Write $\|v\|_A^2 = \int_Q v \cdot Av$ and $\|w\|_{A^{-1}}^2 = \int_Q w \cdot A^{-1} w$.

6.1 Positive periodic rigidity

Lemma 6.1 (Rigidity of a positive mean-zero pair)**.** *Suppose $E = \nabla u$ is a periodic gradient of mean zero, B is periodic and divergence free, and $0 \le J = E \cdot B \in L^1(Q)$. If $R^2 \le KJ$ and $\mathrm{e}^{\tau K} \in L^1(Q)$, then $E = B = 0$ almost everywhere.*

Proof. Equations (4.1) and (4.2) apply on the torus. The normalized periodic potential belongs to $W^{1,q}$ for $q = 2 - 1/(d+1)$, and $uB \in L^1$. Theorem 4.1, applied in coordinate balls, identifies $J = \operatorname{div}(uB)$. These local identities give the global periodic identity, because this is a single periodic potential. Its integral is zero. Positivity makes $J = 0$ almost everywhere, and distortion forces $E = B = 0$. □

6.2 Density in the two actual matrix energy spaces

Define X_A to be the distributional mean-zero periodic gradients in $L^2(A)$. Define $Y_{A^{-1}}$ to be the distributionally divergence-free periodic fields in $L^2(A^{-1})$, with arbitrary means. They are closed Hilbert subspaces: for example

$$\|g\|_1 \le \sqrt{\mu}\, \|g\|_A, \qquad \|g\|_q^2 \le \left(\int_Q K^{q/(2-q)} \,\mathrm{d}x \right)^{(2-q)/q} \|g\|_A^2 \quad (1 < q < 2),$$

and the analogous estimates hold for $Y_{A^{-1}}$. Thus differential constraints and means pass to limits. A mean-zero curl-free periodic L^q field is a periodic gradient, as follows by Fourier approximation and Poincare's inequality.

Proposition 6.2 (Primal and dual energy density)**.** *Smooth periodic gradients are dense in X_A. Smooth periodic divergence-free fields, including constants, are dense in $Y_{A^{-1}}$.*

Proof. Let $\mathcal{H}$ be the A-norm closure of smooth gradients. For $g \in X_A$, project onto $\mathcal{H}$, writing $g = h + r$. The residual r is a mean-zero gradient, and its orthogonality gives $\operatorname{div}(Ar) = 0$. The pair $E = r, B = Ar$ has nonnegative integrable product $r \cdot Ar$. The eigenvalue calculation in Section 5 gives its distortion inequality with the same K. Lemma 6.1 yields $r = 0$.

For the dual statement let $\mathcal{K}$ be the A^{-1}-norm closure of smooth divergence-free fields, and let r be the orthogonal residual of $B \in Y_{A^{-1}}$. Put $E = A^{-1} r$. Orthogonality to constant

test fields gives $\langle E\rangle_Q = 0$. Orthogonality to $(\partial_j\phi)e_i - (\partial_i\phi)e_j$ gives $\partial_j E_i - \partial_i E_j = 0$. Thus E is a mean-zero gradient and $r = AE$ is divergence free. Its product is nonnegative, integrable, and satisfies the same distortion. Lemma 6.1 again yields $r = 0$.

These are projections only in an existence proof. They are not operations assumed available to the computational algorithm. □

The distinction between the matrix energy and the scalar majorant is essential: density is proved in $\int g \cdot Ag$, not in the generally stronger norm $\int K|g|^2$. Scalar-weight density has a substantial history; for example [3, Theorem 4.1] discusses Zhikov-type criteria. That scalar theorem is not substituted for either of the two residual arguments above.

Corollary 6.3 (Admissible pairing)**.** *If $E = \boldsymbol{e} + g$, $g \in X_A$, and $B \in Y_{A^{-1}}$, then*

$$\int_Q E \cdot B \,\mathrm{d}x = \boldsymbol{e} \cdot \langle B\rangle_Q.$$

Proof. The product is integrable by weighted Cauchy–Schwarz. Approximate g by smooth gradients in $L^2(A)$; their pairing with B is zero and passes to the limit. □

6.3 Solvability among all finite-energy weak solutions

For bounded F, both constants and F have finite energies in the appropriate spaces, by (6.1). On $\boldsymbol{e} + X_A$ minimize

$$\mathcal{P}(E) = \int_Q (E \cdot AE - 2F \cdot E) \,\mathrm{d}x.$$

The linear term is continuous by $|\int F \cdot g| \le \|F\|_{A^{-1}}\|g\|_A$. Completing the square in this Hilbert affine space gives a unique minimizer E_*. Its Euler equation against smooth gradients gives

$$B_* = AE_* - F, \qquad \operatorname{div} B_* = 0, \qquad B_* \in Y_{A^{-1}}.$$

The embeddings above and Poincare give $E_* = \boldsymbol{e} + \nabla u_*$, with normalized $u_* \in W^{1,q}$ for every $1 < q < 2$.

If another finite-energy weak solution exists, the difference of its gradient and E_* lies in X_A, and its A-flux is divergence free. Lemma 6.1 eliminates the difference. This proves uniqueness in the maximal finite-energy weak class, not only in the completion first used for existence. It proves the analytic response assertion of Theorem 1.4.

6.4 The hypercircle identity with all pairings justified

For admissible $E \in \boldsymbol{e} + X_A$ and $B \in Y_{A^{-1}}$ define

$$\mathcal{D}(B) = 2\boldsymbol{e} \cdot \langle B\rangle_Q - \int_Q (B + F) \cdot A^{-1}(B + F) \,\mathrm{d}x.$$

This is the classical primal–dual or hypercircle mechanism [16, 19]. Its novelty is not claimed here. We derive its working identity to specify precisely what degeneracy permits.

Theorem 6.4 (Exact primal–dual error)**.** *For every such pair,*

$$\mathcal{P}(E) - \mathcal{D}(B) = \|E - E_*\|_A^2 + \|B - B_*\|_{A^{-1}}^2 \tag{6.2}$$

$$= \int_Q |A^{1/2}E - A^{-1/2}(B + F)|^2 \,\mathrm{d}x. \tag{6.3}$$

In particular $\mathcal{D}(B) \le \mathcal{P}(E_*) \le \mathcal{P}(E)$.

Proof. Expanding $\mathcal{P}$ around E_* leaves $\mathcal{P}(E) - \mathcal{P}(E_*) = \|E - E_*\|_A^2$. The cross term is $2\int(E - E_*) \cdot (AE_* - F) = 2\int(E - E_*) \cdot B_* = 0$ by Corollary 6.3. Similarly

$$\mathcal{D}(B_*) - \mathcal{D}(B) = \|B - B_*\|_{A^{-1}}^2,$$

because $A^{-1}(B_* + F) = E_*$ and $\int E_* \cdot (B - B_*) = \boldsymbol{e} \cdot \langle B - B_* \rangle_Q$. The same pairing gives $\mathcal{P}(E_*) = \mathcal{D}(B_*)$. This proves (6.2). Expand the square in (6.3) and use $\int E \cdot B = \boldsymbol{e} \cdot \langle B \rangle_Q$ to obtain the second identity. Every pairing is a weighted Cauchy–Schwarz pairing, justified by Proposition 6.2; none assumes separate L^2 fields. □

6.5 A rational enclosure with coefficient uncertainty

Suppose bounded rational compatible fields $\widetilde{E}, \widetilde{B}$ are given, together with rational step matrices S, C and

$$\int_Q \|A - S\|_{\mathrm{op}} \,\mathrm{d}x \le \eta, \qquad \int_Q \|A^{-1} - C\|_{\mathrm{op}} \,\mathrm{d}x \le \zeta.$$

Use rational upper bounds $E_\infty \ge \|\widetilde{E}\|_\infty$ and $B_\infty \ge \|\widetilde{B} + F\|_\infty$. The quantities

$$U = \int_Q (\widetilde{E} \cdot S\widetilde{E} - 2F \cdot \widetilde{E}) \,\mathrm{d}x + \eta E_\infty^2, \tag{6.4}$$

$$L = 2\boldsymbol{e} \cdot \langle \widetilde{B} \rangle_Q - \int_Q (\widetilde{B} + F) \cdot C(\widetilde{B} + F) \,\mathrm{d}x - \zeta B_\infty^2 \tag{6.5}$$

are rational and satisfy

$$L \le \mathcal{D}(\widetilde{B}) \le \mathcal{P}(E_*) \le \mathcal{P}(\widetilde{E}) \le U.$$

Hence $U - L \le \epsilon^2$ certifies

$$\|\widetilde{E} - E_*\|_A^2 + \|\widetilde{B} - B_*\|_{A^{-1}}^2 \le \epsilon^2. \tag{6.6}$$

It also gives

$$\|\widetilde{E} - E_*\|_1^2 + \|\widetilde{B} - B_*\|_1^2 \le \mu\epsilon^2.$$

The two different coefficient errors and their sup-norm multipliers are indispensable. Neither approximate matrix need be positive pointwise for these enclosure directions to be valid. The flux in the constitutive law is $B + F = AE$; the certified divergence-free field is B.

7 Represented data and a terminating rational algorithm

An arbitrary measurable coefficient cannot be acquired from finitely many point values. A computational statement therefore needs an information contract. We use approximations in integral norm, whose errors enter the certificate explicitly. The inverse coefficient is constructed from this contract; it is not an additional oracle.

7.1 The input and the finite output

The input consists of d, rational $\tau, M > 0$ satisfying (1.8), rational $\boldsymbol{e} \in \mathbb{Q}^d$, bounded rational dyadic step data F, and a finite program which, for each integer $s \ge 1$, returns an *expanded* rational dyadic step matrix S_s and a rational $\eta_s \le 2^{-s}$ such that

$$\int_Q \|A - S_s\|_{\mathrm{op}} \,\mathrm{d}x \le \eta_s. \tag{7.1}$$

Every box of the stated uniform grid is listed. Symmetrizing S_s preserves the bound, so assume symmetry. Its program runtime $T_A(s)$, expanded output length, grid level, dimension and

coefficient bit lengths are charged. An infinite name and its exponential-moment promise need not be decidable by a finite verifier. The theorem is conditional on this mathematical input contract, as is usual for a represented real input.

Given a positive rational ϵ, the output is a finite rational description of compatible bounded fields $\widetilde{E}, \widetilde{B}$ and an interval $[L, U]$ satisfying (6.4)–(6.6). This approximates continuum fields and their energy; it is not a finite exact encoding of an arbitrary infinite field.

7.2 A rational inverse filter

Lemma 7.1 (Inverse name without inverse access)**.** *For a symmetric matrix S, rational $R \geq 1$, and integer $k \geq 1$, put*

$$T_R(S) = (I + R^2 S^2)^{-1}, \qquad C_{R,k}(S) = R^2 S \sum_{j=1}^{k} T_R(S)^j.$$

If $\int_Q \|A - S\|_{\mathrm{op}} \leq \eta$, then

$$\|A^{-1} - C_{R,k}(S)\|_{L^1(\mathrm{op})} \leq \frac{k(k+3)}{4} R^2 \eta + R 2^{-k} + M(R + 1/\tau)\mathrm{e}^{-\tau R}. \tag{7.2}$$

All operations on a rational step matrix are finite rational operations. The estimate permits noncommuting $A(x)$ and $S(x)$.

Proof. The matrix $I + R^2 S^2 \geq I$, even if S has negative eigenvalues. For the positive true matrix A, the finite geometric sum gives

$$A^{-1} - C_{R,k}(A) = A^{-1} T_R(A)^k.$$

On $\{K \leq R\}$, $A \geq R^{-1} I$, so its norm is at most $R 2^{-k}$. On the complement it is at most K. The distribution bound $|\{K > t\}| \leq M\mathrm{e}^{-\tau t}$ yields

$$\int_{\{K > R\}} K \, \mathrm{d}x \leq M(R + 1/\tau)\mathrm{e}^{-\tau R}.$$

Here is the perturbation argument without a commutativity assumption. For symmetric X set $J_\pm(X) = (X \pm iR^{-1} I)^{-1}$. Their complex operator norms are at most R; the resolvent identity therefore gives $\|J_\pm(X) - J_\pm(Y)\|_{\mathrm{op}} \leq R^2 \|X - Y\|_{\mathrm{op}}$. The identities

$$F_R(X) := R^2 X T_R(X) = \tfrac{1}{2}(J_-(X) + J_+(X)), \qquad T_R(X) = \frac{J_-(X) - J_+(X)}{2iR}$$

imply

$$\mathrm{Lip}(F_R) \leq R^2, \quad \mathrm{Lip}(T_R) \leq R, \quad \|F_R(X)\|_{\mathrm{op}} \leq R/2, \quad \|T_R(X)\|_{\mathrm{op}} \leq 1.$$

Use $C_{R,k} = F_R \sum_{j=0}^{k-1} T_R^j$ and telescope each noncommuting power difference:

$$\|C_{R,k}(X) - C_{R,k}(Y)\|_{\mathrm{op}} \leq R^2 \left(k + \frac{1}{2} \sum_{j=0}^{k-1} j \right) \|X - Y\|_{\mathrm{op}} = \frac{k(k+3)}{4} R^2 \|X - Y\|_{\mathrm{op}}.$$

Integration and the tail estimate prove (7.2). □

An inverse name with error 2^{-n} can be selected entirely rationally. Increase an integer $r \geq 1$ until, with $R = \max\{1, 2r/\tau\}$,

$$M(R + 1/\tau) 2^{-2r} \leq 2^{-n-2}.$$

This bounds the exponential term because $\mathrm{e} \geq 2$ and $\tau R \geq 2r$. Choose $k \geq 1$ and $s \geq 1$ so that

$$R2^{-k} \leq 2^{-n-2}, \qquad \frac{k(k+3)}{4} R^2\, 2^{-s} \leq 2^{-n-1}.$$

Then $C_n = C_{R,k}(S_s)$ has the desired L^1 inverse error. For fixed τ, M, $r, k = O(n + \log(n+1))$ and $s = n + O(\log(n+1))$. These are parameter estimates, not a bound on the coefficient program or on the spatial expansion.

7.3 Finite compatible spaces and exact assembly

Use the periodic dyadic grid of side 2^{-m}, splitting each cube into its $d!$ Freudenthal simplices. Let V_m consist of continuous periodic piecewise affine scalar functions, with one nodal value fixed. The primal affine space is

$$\boldsymbol{e} + \nabla V_m.$$

The dual space consists of constant fields plus

$$b_i = \sum_j \partial_j \Psi_{ij}, \qquad \Psi_{ij} = -\Psi_{ji}, \quad \Psi_{ij} \in V_m.$$

Commuting distributional derivatives give $\operatorname{div} b = 0$, including across all simplex faces. Remove linear dependencies between these rational piecewise constant fields by exact elimination on their arrays. The primal dimension is at most $2^{dm} - 1$, and the dual dimension at most $d + \frac{1}{2} d(d-1) 2^{dm}$.

The spaces are complete for approximation in the two energy norms. Indeed, a smooth periodic gradient is uniformly approximated by the gradient of its nodal affine interpolant. A smooth divergence-free field has its mean plus a smooth antisymmetric periodic potential (obtain it by the nonzero Fourier modes); interpolate that potential instead. Uniform field error tends to zero, so its weighted energy tends to zero because $A, A^{-1} \in L^1$. Combine this with Proposition 6.2.

On a common dyadic simplicial refinement, every field and supplied step coefficient is constant on each simplex. Each integral in the two quadratic objectives is therefore a finite rational sum. For $\widetilde{E} = \boldsymbol{e} + \sum_i c_i g_i$, the primal solve is

$$G^{\mathrm{p}}_{ij} = \int_Q g_i \cdot S_n g_j, \qquad G^{\mathrm{p}} c = -r^{\mathrm{p}}, \quad r^{\mathrm{p}}_i = \int_Q g_i \cdot (S_n \boldsymbol{e} - F).$$

For $\widetilde{B} = \sum_i b_i w_i$, the dual solve is

$$G^{\mathrm{d}}_{ij} = \int_Q w_i \cdot C_n w_j, \qquad G^{\mathrm{d}} b = r^{\mathrm{d}}, \quad r^{\mathrm{d}}_i = \boldsymbol{e} \cdot \langle w_i \rangle_Q - \int_Q w_i \cdot C_n F.$$

Test each Gram matrix for positive definiteness by exact rational LDL^T elimination, checking that every pivot is positive. If it passes, the solve is finite rational elimination; no numerical optimization oracle is involved. A zero-dimensional primal space is allowed and requires no solve. Rational sup-norm bounds for the fields follow, for example, by taking the maximum of the sum of absolute component values on all simplices.

Theorem 7.2 (Uniform finite algorithm)**.** *On every valid input contract, the following algorithm terminates and returns* (6.6)*.*

At stage $n = 1, 2, \ldots$, obtain S_n and construct C_n with inverse error at most 2^{-n}. For each $0 \leq m \leq \lfloor \log_2(n+1) \rfloor$, assemble the two rational forms above. Skip a level at which either positive-definiteness test fails. At any remaining level solve both systems, form L, U using (6.4)–(6.5)*, and stop when $U - L \leq \epsilon^2$.*

Proof. Every finite operation at a stage is effective under the promised coefficient name. For a fixed level m, the exact Gram matrices using A, A^{-1} are positive definite after dependencies are removed. Their approximate entries converge, since the basis fields are bounded and the coefficient errors tend to zero in L^1. Thus eventually both tests pass. Matrix inversion is continuous at these fixed positive-definite matrices, so the rational trial fields converge to the two exact Galerkin optimizers at that level. Their sup norms remain bounded as $n \to \infty$.

Density and Theorem 6.4 imply that there is some finite level m whose exact total squared Galerkin error is less than $\epsilon^2/4$. The schedule visits this level at all sufficiently large stages. At those stages the optimization errors converge to zero, as do both uncertainty terms $\eta_n E_\infty^2$ and $2^{-n} B_\infty^2$. Eventually $U - L < \epsilon^2$. The stopping certificate is sound by Theorem 6.4 and the enclosure directions already proved. This proves termination, without a uniform spatial approximation rate. □

7.4 Paid resources and verification

Let N_* be the first successful stage discovered by this algorithm. It is finite and instance dependent; it is not advice supplied with the input. If ℓ_n is the largest of the trial level, the forcing grid level, and the grid levels returned for S_n, S_{s_n}, a common mesh contains

$$J_n = d!\, 2^{d\ell_n}$$

simplices. Those arrays are expanded; their size is not suppressed.

For precision in the complexity statement, let Z_n be the total binary length of all raw and filtered tables, mesh coordinates, basis arrays and input scalars used across the levels of stage n, plus $n + k_n + r_n + s_n + 1$. The added integers are counted by value, to pay for the finite loops. A conservative implementation using common denominators, determinant-controlled elimination and schoolbook integer arithmetic has cost bounded by

$$C \sum_{n=1}^{N_*} [T_A(n) + T_A(s_n) + (1 + Z_n)^{16}]. \tag{7.3}$$

Appendix B gives the bookkeeping. Caching calls can only reduce this upper bound. Memory is bounded by a fixed polynomial in the largest expanded stage data when stages are discarded after use.

The expression is deliberately in expanded data, not in the length of a short program describing an enormous grid. Neither N_*, T_A, nor those expansions have a uniform polynomial bound here. A certificate checker can verify the finite compatible fields, rational integrals and error allocation without replaying the search. It remains conditional on the coefficient-name and moment promises; it does not decide an infinite mathematical promise.

7.5 Two information and complexity boundaries

Proposition 7.3 (Finite deterministic point access is insufficient)**.** *Finitely many adaptive point values cannot certify the effective response of all measurable coefficients satisfying the same fixed contract $\tau = 1/4, M = 2$, even for scalar multiples of the identity.*

Proof. Answer every query by I. After any finite terminating transcript, the coefficient equal to $2I$ except at the queried points has exactly the same answers as I, but effective tensor $2I$ instead of I. Both obey the stated moment bound. Thus no certified tolerance smaller than half their separation can hold on both inputs. This concerns finite deterministic point information, not the integral-error representation above.

Even restricting to smooth coefficients without a supplied modulus does not give a uniform finite-query guarantee. Put smooth dips to I around the finitely many points and use $2I$

elsewhere, with total support volume $v < 1/100$. For $A = aI$, weighted Cauchy–Schwarz gives $A_* \geq (\int a^{-1})^{-1} I \geq 2(1+v)^{-1} I$. The response remains separated from that of I. □

Proposition 7.4 (A stronger succinct claim would count)**.** *Suppose an algorithm evaluated the effective energy in this coefficient class in time polynomial in a succinct circuit description, dimension, and requested precision bits. Even for $1 \leq A \leq 2$, it would compute #SAT, and in particular imply $P = NP$.*

Proof. For a circuit C on n listed bits take $d = n + 1$ and

$$A_C(x) = \Big[1 + C(\mathbf{1}_{x_2 \geq 1/2}, \ldots, \mathbf{1}_{x_{n+1} \geq 1/2})\Big] I.$$

The coefficient is independent of x_1, so $E = e_1$ is exactly the cell gradient for mean e_1. Its effective energy is

$$e_1 \cdot A_* e_1 = 1 + \#\mathrm{SAT}(C)/2^n.$$

An approximation with error $< 2^{-n-2}$ recovers the integer count by rounding after multiplying by 2^n. Expansion into the 2^n boxes is not polynomial in the succinct description; our algorithm pays for it. This is a conditional complexity implication, not an unconditional separation or a claim about fixed dimension. □

8 Effective response and periodic homogenization

8.1 The effective law and its finite enclosure

Set $F = 0$, and write $E_{\boldsymbol{e}}, B_{\boldsymbol{e}}$ for the cell response. Linearity and uniqueness show that both depend linearly on $\boldsymbol{e}$. Define

$$A_* \boldsymbol{e} = \langle B_{\boldsymbol{e}} \rangle_Q, \qquad \boldsymbol{e} \cdot A_* \boldsymbol{e} = \min_{E \in \boldsymbol{e} + X_A} \int_Q E \cdot AE \, \mathrm{d}x.$$

Corollary 6.3 gives

$$\boldsymbol{e} \cdot A_* f = \int_Q E_{\boldsymbol{e}} \cdot AE_f \, \mathrm{d}x = f \cdot A_* \boldsymbol{e}.$$

Thus A_* is symmetric. The constant trial field gives the upper bound below. For the lower bound, weighted Cauchy–Schwarz applied to $\langle E \rangle_Q = \boldsymbol{e}$ gives $|\boldsymbol{e}|^2 \leq \mu \int E \cdot AE$. Consequently

$$\mu^{-1} I \leq A_* \leq \mu I. \tag{8.1}$$

Solve the d unit-mean cell problems with individual certificate gaps at most $\epsilon_{\mathrm{ten}}^2/(d\mu)$. Let C have columns the rational mean trial fluxes, and put $\widehat{A} = (C + C^T)/2$. Then

$$\|C - A_*\|_{\mathrm{F}}^2 \leq \mu \sum_{j=1}^{d} \|\widetilde{B}_j - B_{e_j}\|_{A^{-1}}^2 \leq \epsilon_{\mathrm{ten}}^2.$$

Symmetrization is a contraction in Frobenius norm. Hence the rational Loewner enclosure is

$$\widehat{A} - \epsilon_{\mathrm{ten}} I \leq A_* \leq \widehat{A} + \epsilon_{\mathrm{ten}} I. \tag{8.2}$$

Every solve and acquisition cost is included. The local logarithmic theorem applies to these now-existing homogeneous cell solutions, with cell energy at most $\mu|\boldsymbol{e}|^2$. It was not used as an existence theorem.

8.2 Small-region relative edits

The energy concentration estimate has a direct response consequence. It controls changes on a small region when the two constitutive laws remain comparable there. This is a deduction from logarithmic regularity and the variational principle, rather than a different construction of an effective medium.

Corollary 8.1 (Relative-edit stability)**.** *Let $A, \widetilde{A}$ both satisfy* (1.8) *with the same τ, M. Suppose they agree outside a measurable $S \subset Q$, $|S| = v \in (0, 1]$, and*

$$\gamma^{-1} A \le \widetilde{A} \le \gamma A, \qquad \gamma \ge 1.$$

Let $E, \widetilde{E}$ be their homogeneous cell gradients with the same mean $\boldsymbol{e}$. For every $0 < \beta < \tau$,

$$\|\widetilde{E} - E\|_{\widetilde{A}}^2 \le (\gamma + \gamma^{-1} - 2) \int_S E \cdot AE \,\mathrm{d}x \le \frac{C(d, \beta, \tau, M, \gamma)|\boldsymbol{e}|^2}{\log^{\beta}(\mathrm{e} + 1/v)}, \tag{8.3}$$

$$\|\widetilde{A}_* - A_*\|_{\mathrm{op}} \le \frac{C(d, \beta, \tau, M, \gamma)}{\log^{\beta}(\mathrm{e} + 1/v)}. \tag{8.4}$$

Proof. First the concentration corollary gives the periodic estimate

$$\int_S E \cdot AE \,\mathrm{d}x \le \frac{C(d, \beta, \tau, M)|\boldsymbol{e}|^2}{\log^{\beta}(\mathrm{e} + 1/v)}. \tag{8.5}$$

Here are the uniform normalization details. For $|\boldsymbol{e}| = 1$ the cell energy is at most μ. Extend the fields periodically and cover the torus by a fixed finite family of inner balls whose concentric doubled balls lie in a bounded union of cells. On every doubled ball both the energy and exponential-moment averages are bounded by constants depending only on d, τ, M. Thus the normalization A_0^2 in Theorem 1.3 is bounded above uniformly. Corollary 4.3 applies to each S intersected with an inner ball. Its relative volume is at most a fixed multiple of v. For small v the logarithms are comparable; for the remaining v use the total energy bound. Zero-energy balls contribute nothing. Summing proves (8.5) for unit mean. Linearity and scaling give arbitrary $\boldsymbol{e}$, including zero. The same argument applies to $\widetilde{A}$.

Relative comparability identifies the two gradient energy spaces. Subtract their Euler equations and test with $w = \widetilde{E} - E$, a mean-zero gradient in both spaces:

$$\|w\|_{\widetilde{A}}^2 = -\int_S w \cdot (\widetilde{A} - A)E \,\mathrm{d}x \le \|w\|_{\widetilde{A}} \left[\int_S E \cdot (\widetilde{A} - A)\widetilde{A}^{-1}(\widetilde{A} - A)E \,\mathrm{d}x\right]^{1/2}.$$

For $C = A^{-1/2}\widetilde{A}A^{-1/2}$, the matrix in this last quadratic form is $A^{1/2}(C + C^{-1} - 2I)A^{1/2}$. Its middle factor is at most $(\gamma + \gamma^{-1} - 2)I$, since $\gamma^{-1}I \le C \le \gamma I$. Cancellation of $\|w\|_{\widetilde{A}}$, with the zero case understood, proves the first inequality in (8.3).

The old minimizer is an admissible trial in the edited medium, so

$$\boldsymbol{e} \cdot (\widetilde{A}_* - A_*)\boldsymbol{e} \le \int_S E \cdot (\widetilde{A} - A)E \,\mathrm{d}x \le (\gamma - 1) \int_S E \cdot AE \,\mathrm{d}x.$$

Interchange the two coefficients for the opposite inequality. Apply (8.5) to each, and take the supremum over unit $\boldsymbol{e}$; symmetry gives (8.4). □

Both moment assumptions and bounded relative comparability are part of the conclusion's scope. A replacement with arbitrarily large contrast on a small set is not covered. Nor does knowing the set S supply the represented coefficient data required by the certified algorithm.

8.3 Periodic averaging and admissible macroscopic domains

We use the following elementary averaging fact: if $g \in L^1(Q)$ is periodic and ϕ is continuous with compact support, then

$$\int \phi(x) g(x/\delta)\,\mathrm{d}x \longrightarrow \left(\int_Q g\right) \int \phi.$$

One proof approximates g in $L^1(Q)$ by trigonometric polynomials. Periodic cell covering bounds the approximation error uniformly on the support of ϕ; each nonzero Fourier mode tends to zero by oscillatory integration. The same covering gives uniform local L^1 bounds for $g(x/\delta)$. We also use weak-* averaging for bounded periodic g, which follows by approximation of the macroscopic test function in L^1.

Let Ω be bounded Lipschitz and $q = 2 - 1/(d+1)$. For each $\delta > 0$ there are two permitted domain conventions:

- the maximal class $\{u \in W_0^{1,q}(\Omega) : \int_\Omega \nabla u \cdot A(x/\delta)\nabla u < \infty\}$;
- the closure of $C_c^\infty(\Omega)$ in that gradient-energy norm.

The latter embeds in the former. The theorem asserts the same limit for either choice, not an unproved equality of the two spaces at each fixed δ. Extend the energy by $+\infty$ outside the selected domain in $L^2(\Omega)$.

Theorem 8.2 (Degenerate periodic variational homogenization)**.** *With either domain convention,*

$$I_\delta(u) = \int_\Omega \nabla u \cdot A(x/\delta)\nabla u\,\mathrm{d}x \quad \overset{\Gamma,L^2}{\longrightarrow} \quad I_0(u) = \begin{cases} \displaystyle\int_\Omega \nabla u \cdot A_* \nabla u\,\mathrm{d}x, & u \in H_0^1(\Omega), \\ +\infty, & \textit{otherwise.} \end{cases}$$

Bounded-energy sequences for $\delta \downarrow 0$ are precompact in strong $L^2(\Omega)$. For $f \in L^2(\Omega)$, the unique minimizers of $I_\delta(u) - 2\int_\Omega fu$ converge in L^2 to the unique minimizer with A_*. This qualitative assertion requires no smoothness of the macroscopic solution. The additional smooth-data rate is stated separately in Theorem 8.3.

Proof. Coercivity and well-defined minimizers. Put $m = q/(2-q) = 2d+1$. Holder gives

$$\|\nabla u\|_q^2 \le \left(\int_\Omega K(x/\delta)^m\,\mathrm{d}x\right)^{1/m} I_\delta(u). \tag{8.6}$$

Periodic covering of a bounded box containing Ω and (6.1) bound the factor uniformly for $0 < \delta \le 1$. Poincare and the compact embedding $W_0^{1,q} \Subset L^2$ give precompactness, because $q > 2d/(d+2)$. For fixed δ, each domain is a closed Hilbert space in the energy norm. For the maximal domain this follows by convergence of gradients in the weighted Hilbert space and in L^q; the zero trace passes through $W^{1,q}$. The loading is continuous by (8.6). Strict positivity of A and the zero trace give strict convexity and a unique minimizer.

Lower bound. Let $u_\delta \to u$ in L^2 with bounded energy. Take smooth bounded periodic divergence-free fields $b_1, \dots, b_d$ and $p \in C_c^\infty(\Omega; \mathbb{R}^d)$, and put

$$b_\delta(x) = \sum_j p_j(x) b_j(x/\delta).$$

The pointwise square inequality yields

$$I_\delta(u_\delta) \ge 2\int_\Omega \nabla u_\delta \cdot b_\delta\,\mathrm{d}x - \int_\Omega b_\delta \cdot A(x/\delta)^{-1} b_\delta\,\mathrm{d}x.$$

The divergence of b_δ is $\sum_j \nabla p_j \cdot b_j(x/\delta)$, with no $1/\delta$ term. Integration by parts is legitimate for the $W^{1,q}$ function and this smooth bounded field. Strong L^2 convergence and bounded periodic averaging pass the linear term to $-2\int u \sum_j \nabla p_j \cdot \langle b_j\rangle_Q$. The quadratic term averages by the L^1 averaging fact, since the b_j are bounded and $A^{-1} \in L^1(Q)$.

Now approximate the exact cell fluxes B_{e_j} by such b_j in $L^2(A^{-1})$, using Proposition 6.2. Their means converge by the L^1 embedding, and their quadratic matrix converges by weighted Cauchy–Schwarz:

$$\langle b_j\rangle_Q \longrightarrow A_* e_j, \qquad \int_Q b_i \cdot A^{-1} b_j \longrightarrow (A_*)_{ij}.$$

Thus, for every smooth compact p,

$$\liminf_{\delta\downarrow 0} I_\delta(u_\delta) \geq 2\langle \nabla u, A_* p\rangle - \int_\Omega p \cdot A_* p \,\mathrm{d}x. \tag{8.7}$$

The first pairing is initially distributional. Its bounded quadratic dual norm and (8.1) imply $\nabla u \in L^2$: rescale any test p, optimize that scalar factor, and extend the resulting bounded linear functional by L^2 duality. Coercivity also gives $u \in W_0^{1,q}$ by weak convergence. Since $u \in H^1$ and Ω is Lipschitz, its trace is zero, so $u \in H_0^1$. Supremizing (8.7) over smooth p, dense in L^2, proves the lower bound. If the liminf is infinite there is nothing to prove.

Recovery for smooth macroscopic functions. Let $u \in C_c^\infty(\Omega)$. Choose smooth periodic scalar corrector trials ϕ_j, and define

$$u_\delta = u + \delta \sum_j \phi_j(x/\delta)\partial_j u.$$

These functions lie in $C_c^\infty(\Omega)$, so are admissible under both domain conventions, and converge to u in L^2. Their gradients are

$$\sum_j (e_j + \nabla\phi_j)(x/\delta)\partial_j u + \delta \sum_j \phi_j(x/\delta)\nabla\partial_j u.$$

The last term has energy $O(\delta^2)$, since the factors are bounded and $A \in L^1(Q)$. Its cross term tends to zero by weighted Cauchy–Schwarz. Averaging the leading term gives the constant quadratic form with entries

$$Q_{ij}^\phi = \int_Q (e_i + \nabla\phi_i) \cdot A(e_j + \nabla\phi_j)\,\mathrm{d}x.$$

By Proposition 6.2, choose $\nabla\phi_j$ converging in $L^2(A)$ to $E_{e_j} - e_j$. Then $Q^\phi \to A_*$, including the cross entries. Choose this trial index increasingly slowly as $\delta \to 0$: for each fixed index the averaging and remainder estimates hold below a sufficiently small positive threshold; take a decreasing sequence of such thresholds and use that index between successive ones. This diagonal sequence has the required upper limit.

General recovery and minimizers. Approximate any $u \in H_0^1$ by C_c^∞ in H^1; (8.1) gives convergence of the limit energies. A second diagonalization gives recovery for u. Outside H_0^1 the upper-limit requirement is automatic. This proves Gamma convergence with equicoercivity. The loading is continuous in strong L^2. Its bounded minimizing sequences have bounded energy by (8.6) and Young's inequality. The lower and recovery bounds imply that every cluster point minimizes the limit functional. Its strict convexity gives uniqueness and hence convergence of the whole family of minimizers. □

This proof is a degenerate instance of classical periodic variational homogenization [8, Chapter 14]. It does not import a uniform ellipticity assumption into (1.8), or establish a stochastic or nonperiodic theorem.

8.4 A finite-scale estimate for smooth macroscopic data

Qualitative homogenization permits arbitrary L^2 loadings on a Lipschitz domain. A rate requires more information. Here we impose smoothness on the homogenized solution, while leaving the microscopic coefficient in exactly the same exponential-moment class. The first-order corrector, flux potential and boundary cutoff are established quantitative homogenization tools. For comparison, Griso's uniformly elliptic result [11, Theorem 4.1 in the cited preprint] gives a square-root-scale corrector estimate on $C^{1,1}$ domains with L^2 data. Degenerate quantitative theory also exists in stochastic settings, with distinct moment and ensemble assumptions [7, Definition 1.1 and Section 2]. We prove the particular weighted estimates and variational admissibility needed under (1.8); these other theorems are not imported into this coefficient class.

Let Ω be a bounded C^2 domain and write

$$A_\delta(x) = A(x/\delta), \qquad \|g\|_{A_\delta,\Omega}^2 = \int_\Omega g \cdot A_\delta g \, \mathrm{d}x.$$

Let $E_i = e_i + \nabla\phi_i$, $B_i = AE_i$ be the unit-mean cell solutions, with $\langle\phi_i\rangle_Q = 0$. Choose periodic antisymmetric flux potentials with zero mean so that

$$b_i = B_i - A_* e_i, \qquad \sum_k \partial_{y_k}\sigma_{i,jk} = (b_i)_j, \qquad \sigma_{i,jk} = -\sigma_{i,kj}. \tag{8.8}$$

The next lemma constructs them and controls their weighted norms. For any periodic object g, abbreviate $g^\delta(x) = g(x/\delta)$. Choose $\eta_\delta \in C_c^\infty(\Omega)$ with

$$0 \le \eta_\delta \le 1, \quad \eta_\delta = 0 \text{ if } \operatorname{dist}(x,\partial\Omega) < \delta, \quad \eta_\delta = 1 \text{ if } \operatorname{dist}(x,\partial\Omega) > 3\delta, \quad |\nabla\eta_\delta| \le C_\Omega/\delta.$$

Such cutoffs exist for $0 < \delta \le \delta_0(\Omega)$.

Theorem 8.3 (Smooth-data square-root rate)**.** *Assume* (1.8). *Suppose* $u_0 \in C^2(\overline{\Omega}) \cap H_0^1(\Omega)$, *and set*

$$f = -\operatorname{div}(A_*\nabla u_0).$$

For either of the two macroscopic domain conventions in Theorem 8.2, let u_δ *be the unique minimizer of* $I_\delta(u) - 2\int_\Omega fu$. *Define*

$$v_\delta = u_0 + \delta\eta_\delta \sum_{i=1}^d \phi_i^\delta \, \partial_i u_0, \qquad \mathcal{E}_\delta = \sum_{i=1}^d E_i^\delta \, \partial_i u_0. \tag{8.9}$$

There are $C = C(d,\tau,M,\Omega) < \infty$ *and* $0 < \delta_0 = \delta_0(\Omega) \le 1$, *independent of the microscopic coefficient beyond its stated moment bound, such that for* $0 < \delta \le \delta_0$,

$$\|u_\delta - u_0\|_{L^2(\Omega)} \le C\sqrt{\delta}\,\|u_0\|_{W^{2,\infty}(\Omega)}, \tag{8.10}$$

$$\|\nabla u_\delta - \nabla v_\delta\|_{A_\delta,\Omega} + \|\nabla u_\delta - \mathcal{E}_\delta\|_{A_\delta,\Omega} \le C\left(\sqrt{\delta}\,\|\nabla u_0\|_\infty + \delta\|D^2 u_0\|_\infty\right). \tag{8.11}$$

The trial v_δ *belongs to both admitted domains. No equality of these domains at a fixed* δ *is assumed.*

The hypothesis is on the actual macroscopic solution, not merely on its loading: $f \in L^2$ by itself does not supply the stated C^2 and $W^{2,\infty}$ bounds. The corrected-gradient estimate is in the matrix energy; it is not an unweighted L^2 gradient estimate. Neither rate is asserted optimal.

Lemma 8.4 (Weighted cell potentials)**.** *Put $q = 2 - 1/(d+1)$, $m = 2d+1$, and $b_0 = 2m/(m-1) = 2 + 1/d$. The potentials in* (8.8) *can be chosen with*

$$\sum_i \left(\|\phi_i\|_{W^{1,q}(Q)} + \|\sigma_i\|_{W^{1,q}(Q)} \right) \le C(d, \tau, M).$$

Moreover

$$\sum_i \int_Q \left[E_i \cdot AE_i + \nabla\phi_i \cdot A\nabla\phi_i + K(|\phi_i|^2 + |\sigma_i|^2) \right] \mathrm{d}x \le C(d, \tau, M). \tag{8.12}$$

Proof. The constant trial fields give $\int_Q E_i \cdot AE_i \le \mu = M/\tau$. The same number is the A^{-1} energy of B_i. Weighted Holder, with exponent $m = q/(2-q)$, therefore bounds E_i, B_i in L^q, using $\int K^m \le m! M/\tau^m$. Subtract their means and use Poincare to obtain $\phi_i \in W^{1,q}$ and $b_i \in L^q$ with controlled norms.

For an explicit periodic flux potential let h_i be the mean-zero vector solution of $\Delta h_i = b_i$, and set $\sigma_{i,jk} = \partial_k (h_i)_j - \partial_j (h_i)_k$. Fourier modes give $\operatorname{div} h_i = 0$ and $\operatorname{div} \sigma_i = b_i$, as well as antisymmetry. The operators $\partial_l \partial_k \Delta^{-1}$ are the component multipliers of the gradient projection P. Their L^q bounds follow, for example, from $P = (I + H)/2$, already proved. Applied to the derivatives of σ_i, these bounds and Poincare prove its $W^{1,q}$ bound. This constructs the potential without any elliptic regularity assumption on A.

The Sobolev exponent satisfies

$$q^* = \frac{dq}{d-q} > b_0 = \frac{2m}{m-1}.$$

Indeed $q = (2d+1)/(d+1)$ and $q^*/b_0 = d^2/(d^2 - d - 1) > 1$. Holder consequently gives

$$\int_Q K(|\phi_i|^2 + |\sigma_i|^2) \le \|K\|_m (\|\phi_i\|_{b_0}^2 + \|\sigma_i\|_{b_0}^2) \le C.$$

Finally $\nabla\phi_i \cdot A\nabla\phi_i \le 2E_i \cdot AE_i + 2e_i \cdot Ae_i$. The integral of the last term is at most 2μ. These estimates prove (8.12). □

Lemma 8.5 (Whole-cell bounds at the boundary)**.** *For each fixed $c > 0$, every nonnegative periodic $g \in L^1(Q)$ satisfies, for $0 < \delta \le \delta_0(\Omega, c)$,*

$$\int_\Omega g^\delta \, \mathrm{d}x \le C_\Omega \int_Q g \, \mathrm{d}x, \tag{8.13}$$

$$\int_{\Omega \cap \{\operatorname{dist}(x, \partial\Omega) < c\delta\}} g^\delta \, \mathrm{d}x \le C_{\Omega,c} \delta \int_Q g \, \mathrm{d}x. \tag{8.14}$$

Proof. Cover the integration set by all grid cubes $\delta(z + Q)$ that meet it. Each contributes exactly $\delta^d \int_Q g$. For the full domain, their disjoint union lies in a fixed bounded neighbourhood, hence has bounded volume. For the strip, this union lies within distance $(c + \sqrt{d})\delta$ of the boundary, on either side. A compact C^2 boundary has a tubular neighbourhood of volume at most $C_{\Omega,c}\delta$. Multiplying the number of covering cubes by their common cell integral proves both statements. The proof uses an integral over whole cells, not a pointwise bound on g. □

Lemma 8.6 (The cutoff corrector and its domain)**.** *The trial in* (8.9) *belongs to the smooth-energy completion, and hence to the maximal domain. With $d_\delta = \nabla v_\delta - \mathcal{E}_\delta$,*

$$d_\delta = (\eta_\delta - 1) \sum_i (\nabla\phi_i)^\delta \, \partial_i u_0 + \delta \sum_i \phi_i^\delta (\eta_\delta \nabla \partial_i u_0 + \partial_i u_0 \nabla \eta_\delta), \tag{8.15}$$

$$\|d_\delta\|_{A_\delta, \Omega} \le C \left(\sqrt{\delta} \, \|\nabla u_0\|_\infty + \delta \|D^2 u_0\|_\infty \right), \tag{8.16}$$

$$\|v_\delta - u_0\|_2 \le C\delta \|\nabla u_0\|_\infty. \tag{8.17}$$

Proof. The weak product rule gives (8.15). The first term is supported in the 3δ boundary strip. Apply (8.14) with $g = \sum_i \nabla\phi_i \cdot A\nabla\phi_i$. Its squared energy is at most $C\delta\|\nabla u_0\|_\infty^2$. For the term with $\nabla\eta_\delta$, use the same bound with $g = \sum_i K|\phi_i|^2$ and $\delta|\nabla\eta_\delta| \le C$. Its squared energy obeys the same estimate. For the remaining term use (8.13); its squared energy is at most $C\delta^2\|D^2u_0\|_\infty^2$. Lemma 8.4 controls all cell integrals, which proves (8.16). The full-cell bound for $\sum_i |\phi_i|^2$ proves (8.17).

It remains to justify the trial's variational domain. By Proposition 6.2, take smooth mean-zero periodic $\phi_i^{(n)}$ with $\nabla\phi_i^{(n)} \to \nabla\phi_i$ in $L^2(A)$. Weighted Holder and Poincare give convergence in $W^{1,q}$. The embedding used above then gives convergence in L^{b_0}, and hence in $L^2(K)$. For each fixed δ, whole-cell covering and the bounded macroscopic multipliers in (8.15) imply energy convergence of

$$\delta\eta_\delta \sum_i (\phi_i^{(n)})^\delta \partial_i u_0 \quad \text{to} \quad \delta\eta_\delta \sum_i \phi_i^\delta \partial_i u_0.$$

Each approximant is C^1 and compactly supported in Ω. Mollification approximates its gradient uniformly, so also in the A_δ energy, since $A_\delta \in L^1(\Omega)$. It therefore belongs to the completion of C_c^∞.

The uncorrected u_0 also belongs to this completion. Cut it off at boundary distance $h \downarrow 0$. In a boundary tube, $|u_0(x)| \le C_\Omega \operatorname{dist}(x, \partial\Omega)\|\nabla u_0\|_\infty$. Thus the error gradients are uniformly bounded and supported in a shrinking strip. Absolute continuity of the integral of K^δ makes their energy tend to zero. Each compactly supported C^1 cutoff can again be mollified with uniform gradient convergence. This proves the claimed membership, without identifying the two domain conventions. □

Lemma 8.7 (Flux-potential pairing in the maximal domain)**.** *If $w \in W_0^{1,q}(\Omega)$ has finite A_δ energy, then, for $p_i = \partial_i u_0$,*

$$\int_\Omega b_i^\delta p_i \cdot \nabla w \,\mathrm{d}x = -\delta \sum_{j,k} \int_\Omega \sigma_{i,jk}^\delta (\partial_k p_i)(\partial_j w)\,\mathrm{d}x. \tag{8.18}$$

Both are weighted energy pairings, and the right-hand side has absolute value at most $C\delta\|D^2u_0\|_\infty\|\nabla w\|_{A_\delta,\Omega}$.

Proof. For smooth compact w, use $(b_i^\delta)_j = \delta\sum_k \partial_k\sigma_{i,jk}^\delta$ and integrate by parts. The terms $\sigma_{i,jk}^\delta p_i \partial_k\partial_j w$ cancel by antisymmetry. Multiplication by the C^1 function p_i is justified by approximation, giving (8.18).

To pass to the maximal domain without presuming Dirichlet density, extend w by zero to a containing cube $\mathcal{C}_\delta$ of side $L_\delta = N\delta$, N an integer. Choose these cubes with a fixed positive margin around $\overline{\Omega}$, and uniformly bounded side lengths. The zero extension is periodic $W^{1,q}$ on the cube and has mean-zero gradient of finite energy. The coefficient A_δ is periodic there. After rescaling the cube to unit size, its normalized exponential moment is at most M, because the cube contains an integer number of periods in each coordinate. Proposition 6.2 therefore approximates this gradient by smooth periodic gradients in its actual matrix energy.

Extend p_i to a compactly supported $W^{1,\infty}$ function on the cube. The C^2 boundary permits such an extension; only existence is needed for this density passage. Apply the smooth identity on the cube with this extension. It is valid for a Lipschitz multiplier by mollification: the multipliers converge uniformly, their derivatives are uniformly bounded and converge almost everywhere, and the weighted integrable functions below give dominated convergence. Both sides are continuous in the gradient energy. In fact

$$\int_Q b_i \cdot A^{-1} b_i \le 2\int_Q B_i \cdot A^{-1} B_i + 2|A_* e_i|^2 \int_Q K < \infty, \qquad \int_Q K|\sigma_i|^2 < \infty.$$

Whole-cell covering gives their counterparts on the cube. Weighted Cauchy–Schwarz therefore passes the identity to the zero-extended gradient. That gradient vanishes outside Ω, so the resulting identity is (8.18).

Finally apply weighted Cauchy–Schwarz on Ω to its right-hand side, with $A_\delta^{-1} \le K^\delta I$, and use (8.13) and (8.12). This gives the asserted bound with the actual interior derivatives $\partial_k p_i = \partial_k \partial_i u_0$. □

Proof of Theorem 8.3. For every maximal-domain w, ordinary zero-trace Sobolev integration by parts against the C^1 flux $A_*\nabla u_0$ gives $\int_\Omega A_*\nabla u_0 \cdot \nabla w = \int_\Omega f w$. For example approximate w in $W_0^{1,q}$; the flux and its divergence are bounded. The corrected gradient satisfies $A_\delta \mathcal{E}_\delta = \sum_i B_i^\delta \partial_i u_0$. Consequently Lemmas 8.6 and 8.7 give

$$\begin{aligned}\left|\int_\Omega A_\delta \nabla v_\delta \cdot \nabla w \,\mathrm{d}x - \int_\Omega f w \,\mathrm{d}x\right| &\le \left(\|d_\delta\|_{A_\delta,\Omega} + C\delta \|D^2 u_0\|_\infty\right) \|\nabla w\|_{A_\delta,\Omega} \\ &\le C\left(\sqrt{\delta}\,\|\nabla u_0\|_\infty + \delta \|D^2 u_0\|_\infty\right) \|\nabla w\|_{A_\delta,\Omega}.\end{aligned}$$

The Euler equation for u_δ holds for every test in its chosen Hilbert domain. The trial v_δ belongs to either domain, so take $w = u_\delta - v_\delta$. Subtracting that equation bounds $\|\nabla u_\delta - \nabla v_\delta\|_{A_\delta,\Omega}$ by the final coefficient on the right. The triangle inequality and (8.16) prove (8.11).

Equation (8.6), periodic m-moment covering, Poincare and $W_0^{1,q}(\Omega) \hookrightarrow L^2(\Omega)$ give

$$\|u_\delta - v_\delta\|_2 \le C(d,\tau,M,\Omega)\|\nabla u_\delta - \nabla v_\delta\|_{A_\delta,\Omega}.$$

Combine this with (8.17) and $\delta \le 1$ to obtain (8.10). □

8.5 An infinite laminate outside A_2

On Q in dimension two, set

$$I_j = (2^{-j}, 2^{1-j}], \quad a(x_1) = \begin{cases} j, & x_1 \in I_j,\ j \text{ odd}, \\ 1/j, & x_1 \in I_j,\ j \text{ even}, \end{cases} \qquad A(x) = a(x_1) I.$$

Null-set values at endpoints may be assigned arbitrarily. Both A and A^{-1} are essentially unbounded. Here $K = j$ on I_j, so

$$\int_Q \mathrm{e}^{K/4}\,\mathrm{d}x = \sum_{j\ge 1} 2^{-j}\mathrm{e}^{j/4} < 2.$$

Indeed $\mathrm{e}^{1/4} < 4/3$, for example by integrating $1/x$ on $[1, 4/3]$. Thus $\tau = 1/4, M = 2, \mu = 8$ is a valid rational contract. At an interface between odd j and even $j+1$, a sufficiently small centered ball has equal halves with values j and $1/(j+1)$. Its product of the averages of a and a^{-1} is at least $j(j+1)/4$. Hence this scalar weight is not Muckenhoupt A_2.

Let a_J agree with a on the first J strips and equal one on the remaining tail. The explicit coefficient-name error is

$$\int_Q |a - a_J|\,\mathrm{d}x \le \sum_{j>J} (j+1)2^{-j} = (J+3)2^{-J}.$$

Choosing J by a rational stopping test gives (7.1). Expanding it onto a uniform dyadic grid is expensive and is charged in the universal algorithm. For this special example, a separate finite certificate integrates the explicitly listed strips directly, paying $O(J)$ strip data instead of pretending to expand a huge grid.

For mean e_1, $E = (h/a, 0)$, $B = (h, 0)$, $h = (\int_Q a^{-1})^{-1}$. For mean e_2, $E = (0, 1)$, $B = (0, a)$. The constraints and finite energies are immediate, and uniqueness identifies these as the cell solutions. Their means give

$$A_* = \operatorname{diag}\left(\frac{1}{8/9 + \frac{1}{2}\log 3}, \frac{10}{9} + \tfrac{1}{2}\log(4/3)\right). \tag{8.19}$$

For clarity, the even integer sum is $\sum_{k\geq 1} 2k\, 4^{-k} = 8/9$, and the odd one is $10/9$. The reciprocal odd sum is $\sum_{j \text{ odd}} 2^{-j}/j = \frac{1}{2}\log 3$, while the reciprocal even sum is $\frac{1}{2}\log(4/3)$. These follow from the geometric series and $\sum_{k\geq 1} z^k/k = -\log(1-z)$, obtained by integration on $|z| < 1$. Formula (8.19) is an identity for comparison, not a numerical logarithm oracle.

8.6 Finite rational laminate trials

Put $t_J = 2^{-J}$, $h_J = (\int_Q a_J^{-1})^{-1}$, and take

$$\widetilde{E}_1 = (h_J/a_J, 0), \quad \widetilde{B}_1 = (h_J, 0), \qquad \widetilde{E}_2 = (0, 1), \quad \widetilde{B}_2 = (0, a_J).$$

Their means and differential constraints are exact. Let O_J, N_J be respectively the odd and even integer tails $\sum_{j>J,\ \text{parity}} j2^{-j}$; let w_o, w_e be their corresponding strip volumes. All four are rational geometric sums. Then rational upper bounds for the two true tails are

$$a_{\text{tail}} \leq U_a := O_J + \frac{w_e}{J+1}, \qquad (a^{-1})_{\text{tail}} \leq U_b := N_J + \frac{w_o}{J+1}.$$

Writing $a_{\text{pre}} = \sum_{j\leq J} 2^{-j} a_j$, primal–dual evaluation yields

$$\begin{aligned} L_1 &= h_J - h_J^2(U_b - t_J), & U_1 &= h_J + h_J^2(U_a - t_J), \\ L_2 &= a_{\text{pre}} + 2t_J - U_b, & U_2 &= a_{\text{pre}} + U_a. \end{aligned}$$

These enclose the two exact minimum energies. Their widths are exactly

$$U_1 - L_1 = h_J^2 T_J, \qquad U_2 - L_2 = T_J, \qquad T_J = \left[J + \frac{1}{J+1}\right] 2^{-J}. \tag{8.20}$$

Indeed $O_J + N_J = (J+2)2^{-J}$ and $w_o + w_e = t_J$. This also proves the bound for the full tail mismatch $\sum_{j>J} 2^{-j}(a_j + a_j^{-1} - 2)$.

The supplied new script evaluates these fractions at $J = 50$, without logarithms or floating-point decisions. The widths are less than $2.148 \cdot 10^{-14}$ and $4.443 \cdot 10^{-14}$, respectively. Their sum, multiplied by $\mu = 8$, is less than 10^{-12}. Thus the rational matrix of mean trial fluxes has certified operator error less than 10^{-6}. Exact fractions and the actually measured numerator/denominator lengths are saved with the source. This finite example illustrates the certificate; the universal termination assertion is proved separately.

9 What the results cover and what remains open

The main analytic construction is the real-vector calibration in arbitrary dimension. It gives the exact reflection norm, the maximal uniform strong scalar interval and the sharp local weak upper endpoint. The weak estimate retains the threshold and integrates a tail differential inequality; the logarithmic theorem instead uses damped threshold weights. The nonlinear uniformly convex result uses the same reflection through a pointwise reflected inverse. The degenerate response and computation theorems are distinct applications: their extra work is energy-space admissibility, represented coefficient access and finite certification.

Several boundaries are substantive. The sharp interval is uniform over a class, not an obstruction for every individual coefficient. At its upper endpoint, homogeneous energy solutions still have the local weak distribution bound, for both gradient and normalized flux. The radial example excludes larger weak exponents and every finite Lorentz index there. A global forced endpoint theorem is not established. The local logarithmic theorem concerns existing pairs with an integrable positive energy product. Its endpoint examples show both that $\beta < \tau$ is necessary in general and that exponential distortion is compatible with the absence of every superquadratic field moment. No assertion about general elliptic systems, higher-degree sharp Hodge norms, or arbitrary boundary regularity has been made.

The response theorem allows unbounded coefficients and inverse coefficients, but retains scalar linear conduction, symmetry, periodicity, fixed units and a supplied exponential moment. Small-region edits have a logarithmic response modulus when both media satisfy the stated moments and remain relatively comparable. Qualitative homogenization holds for L^2 loading on bounded Lipschitz domains. The square-root finite-scale estimate additionally requires a C^2 domain and an actual $C^2(\overline{\Omega}) \cap H_0^1$ homogenized solution, controlled in $W^{2,\infty}$. Its corrected-gradient error is measured in the matrix energy. General-data or optimal rates and stochastic extensions are not proved. The rational solver is complete for its information representation, while retaining potentially large acquisition and spatial costs. The point-access and counting examples explain why stronger statements require stronger hypotheses.

Concrete further questions are whether useful spatial approximation rates for the finite approximation of cell responses can be derived for natural subclasses of the exponential-moment class, and whether these yield certified complexity bounds that improve on expanded finite-dimensional solution at matched input cost. Another question is which nonsymmetric constitutive laws admit a replacement for the positive matrix-energy rigidity used in density. Neither question is settled by the present certificates. The logarithmic endpoint itself cannot be added without an additional assumption, as the explicit radial family demonstrates.

Acknowledgements and AI use

The mathematical ideas, constructions, proofs, and manuscript were developed by GPT-6 Astra, with additional AI assistance for review. Human input was minimal and consisted primarily of setting the research objectives and providing occasional feedback.

A Finite Fourier certificates and bit costs

The finite Fourier construction uses exact complex rational coefficients with conjugate symmetry; the resulting physical fields are real. For the example in Section 3, $\vartheta = 9/10$, $C_4 = 3$, and $\|P\|_4 \le 2$. Consequently a rational bound r_4 for the residual L^4 norm certifies gradient error at most $60r_4$. The sum, over all components and Fourier modes, of the absolute values of real and imaginary coefficient parts is such a bound: triangle inequality bounds the pointwise Euclidean norm by this sum.

The new reconstruction script takes ten exact Picard steps and obtains 40 nonzero gradient modes and eight nonzero residual modes. Its certified gradient error is

$$\frac{2980987023573}{1627930625000000} < 0.001832 < \frac{1}{256}. \tag{A.1}$$

Here the residual is $Hf_{10} - Mf_{10} - h$. For the a priori cap, the coefficient bound for h is two, so (3.13) gives $120(9/10)^N \le 1/256$ first at $N = 99$. The ten-step residual stops earlier, while the cap proves termination regardless of the behaviour of that optional coefficient bound.

The script checks $M^2 = k^2 I$ by finite convolution, the two Cayley/forcing identities, conjugate symmetry, and $\operatorname{curl}(Pf_{10}) = 0$, $\operatorname{div}((I-P)f_{10}) = 0$ coefficientwise. It also checks the exact constitutive defect identity

$$A(Pf_{10}) - (I-P)f_{10} - F = \tfrac{1}{2}(I+A)(Hf_{10} - Mf_{10} - h).$$

Thus the two finite fields have exact differential constraints; their constitutive equation is approximate, with the stated certified residual. These different facts are not conflated. The zero Fourier mode of H, and the case $k = 0$, are checked as well. The saved JSON contains the actual fields and fractions.

A.1 A bound for the finite Fourier procedure

For a general represented Fourier instance take rational p, k, ϵ, with $kC_p < 1$, and rational coefficients for M, h. Include these scalars and frequency indices in an input bit-length bound L. Let S bound the number of frequency indices in either input, and let D_M, D_h bound their coordinate degrees. Assume a valid supplied bound $\|M\|_\infty \le k$; it can, for example, follow from a rational coefficient bound, or from symmetry and an exact identity $M^2 = k^2 I$. The latter verification requires only a finite convolution. Let N be the cap chosen from (3.13). When $k = 0$ one step is exact.

Set

$$R_f = \max\{1, D_h + (N+1)D_M, 2D_M\}, \quad V_f = (2R_f+1)^d, \quad b_f = \left\lceil \log_2(2 + dR_f^2) \right\rceil.$$

The box includes every iterate, reconstruction, residual, and the optional identity verification. Let Q_0 be the product of all input denominators, and Q_H the product of $|m|^2$ over all nonzero indices in the box. Then $\log_2 Q_0 \le L$, $\log_2 Q_H \le V_f b_f$. The coefficients of f_j have denominators dividing $(Q_0 Q_H)^j$, by induction on the Picard recurrence. One extra factor accommodates Pf_N and the residual. Convolution increases coefficient magnitudes by at most a factor exponential in $L + \log_2(d(S+1)+2)$ per iteration. Thus common-denominator numerators and denominators have bit lengths bounded by

$$B_f = O((N+2)[L + V_f b_f + \log_2(d(S+1)+2)]).$$

Constructing the multiplier table and common denominators, checking the optional matrix identity, and applying finite convolutions gives the conservative schoolbook bound

$$O\Big([L^2 + d^3 S^2 + dV_f^2 + (N+1)d^2 V_f (S+1)]B_f^2\Big).$$

Using reduced fractions with elementary gcd arithmetic permits the looser B_f^3 bound. Output memory is $O(dV_f B_f)$; keeping the multiplier and matrix tables costs $O(d^2 V_f B_f + d^2 S B_f + L)$. All estimates pay for the expanded frequency box. They are not polynomial bounds in variable d, binary-encoded frequency radius, or inverse contraction margin. For fixed d, bounded degrees and a fixed contraction margin, the stated procedure is polynomial in encoded input and accuracy bits.

B Rational assembly and the response cost bound

We give further details supporting (7.3), so that the finite solves do not conceal a continuum or real-arithmetic oracle. Integers use binary encoding and ordinary exact arithmetic.

B.1 Inverse filtering on one box

If the entries of a symmetric d-by-d rational S have at most B bits, clear all their denominators using a product with at most d^2B bits. Including the numerator and denominator of R, the integer matrix representing $I + R^2S^2$ has bit length polynomial in $d, B, \mathrm{bits}(R)$. Its determinant is positive. Determinant and adjugate formulas, or fraction-free elimination, compute its inverse with bit lengths bounded by a factor $O(d)$ times those of the cleared entries and $\log(d+1)$, by the elementary determinant bound. Each additional finite power of $T_R(S)$ increases a common-denominator bit bound additively. Thus k powers and their sum cost a polynomial in $d, k, B, \mathrm{bits}(R)$. No eigenvalue approximation is performed.

B.2 Finite optimization and the exponent in the bound

At a fixed stage all grids, basis fields and rational coefficients are explicitly expanded. A common denominator of these supplied rationals has bit length at most their total binary length Z_n. Each assembled Gram or loading entry is a sum of products of a bounded number of table entries, basis values and simplex volumes. The latter are $2^{-d\ell_n}/d!$; their integer descriptions are included in the expanded geometry. The resulting rational entry bit lengths are bounded by $C(1+Z_n)^2$, with a universal constant. There are at most Z_n rows and columns and at most Z_n visited levels and boxes, after harmless zero-size conventions.

To make exact elimination concrete, clear the common denominator of the assembled matrix and right-hand side. Its bit length is at most $C(1+Z_n)^3$. Every elimination pivot and solution coordinate is a quotient of minors of this integer system. The determinant expansion bounds their bit lengths by $C(1+Z_n)^4$. Fraction-free elimination uses $O((1+Z_n)^3)$ integer operations per system, on integers with these determinant-controlled lengths. Schoolbook multiplication, division and gcd have cost at most cubic in that length. These choices put the finite assemblies, solves, compatibility checks and final integrals below $C(1+Z_n)^{16}$, including the at most Z_n levels.

For clarity about inverse-filter intermediates, its raw box matrices, loop counts and filtered outputs are all included in Z_n. Using the denominator product for a box inverse and its finite powers bounds intermediate lengths polynomially in this expanded record. One can use a bound of degree four in $1+Z_n$ even without using the already-recorded filtered-output length: the determinant cost adds a factor d, and common denominators of the powers add a factor k. There are at most Z_n^2 box–power iterations, each with at most $O(Z_n^3)$ scalar matrix operations. Quadratic schoolbook arithmetic, with determinant-controlled exact divisions, keeps this filtering cost below the displayed exponent. Parameter selection uses rational comparisons on loops of lengths r_n, k_n, s_n, already counted by value.

Finally the name program is called at n and s_n, and its actual runtimes are added separately. Expanding its output is already part of that runtime or the charged stage arrays, never a unit-cost operation. This proves (7.3). The bound can be made much smaller for particular representations; it is stated to expose the paid resources, not to claim numerical efficiency.

C Exact geometric tails for the laminate

Let $j_0 > J$ be the first index of a chosen parity. The tail volume and integer tail are

$$\sum_{\nu\ge 0} 2^{-j_0-2\nu} = \frac{4}{3}\,2^{-j_0}, \qquad \sum_{\nu\ge 0}(j_0+2\nu)2^{-j_0-2\nu} = 2^{-j_0}\left(\frac{4j_0}{3}+\frac{8}{9}\right).$$

The reciprocal tail is bounded above by the first expression divided by $J+1$, because every denominator is larger than J. These are the only infinite-series bounds used to construct L_1, U_1, L_2, U_2; all finite prefixes are exact fractions. The new certificate records also include

the squared tensor-error bound $8[(U_1 - L_1) + (U_2 - L_2)]$. At $J = 50$, the largest numerator or denominator among all recorded fractions, including that combined bound, has 280 bits. This measured size refers to the supplied reconstruction, not to a universal bit bound for degenerate coefficients.

D Dependencies and proof boundaries

The central analytic dependencies are explicit: Lemma 2.1 adapts a known scalar construction and proves the needed all-dimensional version; Proposition 2.3 supplies the common calibration; Lemma 2.5 and duality prove the reflection norm; the Cayley calculation proves the uniform scalar and nonlinear inversion results. The local weak endpoint uses the open-interval inverse, scalar and antisymmetric cutoff potentials, the calibration and its tail differential inequality. Its products are ordinary L^2 products; it does not use the critical-product theorem. The logarithmic theorem uses the calibration, the stated critical-product import, regularized potentials, and the damped absorption argument.

For response, Lemma 6.1 uses the critical-product theorem but does not need the full logarithmic improvement. It implies both density statements, hence existence in the maximal finite-energy class and the admissible hypercircle pairing. The rational inverse filter and finite-space search then give certified termination. Homogenization uses the two energy-density statements, periodic averaging and compactness. The quantitative smooth-data theorem further proves K-weighted bounds on the cell and flux potentials, whole-cell boundary-strip control and the corrected trial's membership in both variational domains. Periodic density on a containing cube justifies the residual for maximal-domain tests. The relative-edit corollary combines the logarithmic concentration estimate with the variational principle and bounded relative comparability. Neither the finite certificate scripts nor the computational termination theorem is a premise for an analytic estimate.

The power strong-endpoint and finite-Lorentz-index obstructions, and the logarithmic endpoint and no-power-gain constructions, are counterexamples to the stated stronger conclusions, with their assumptions checked directly. The succinct counting construction is instead a conditional complexity implication; it is not an unconditional hardness separation.

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